

SQANTI3 filter report

Total Genes: 667

Total Transcripts: 3925

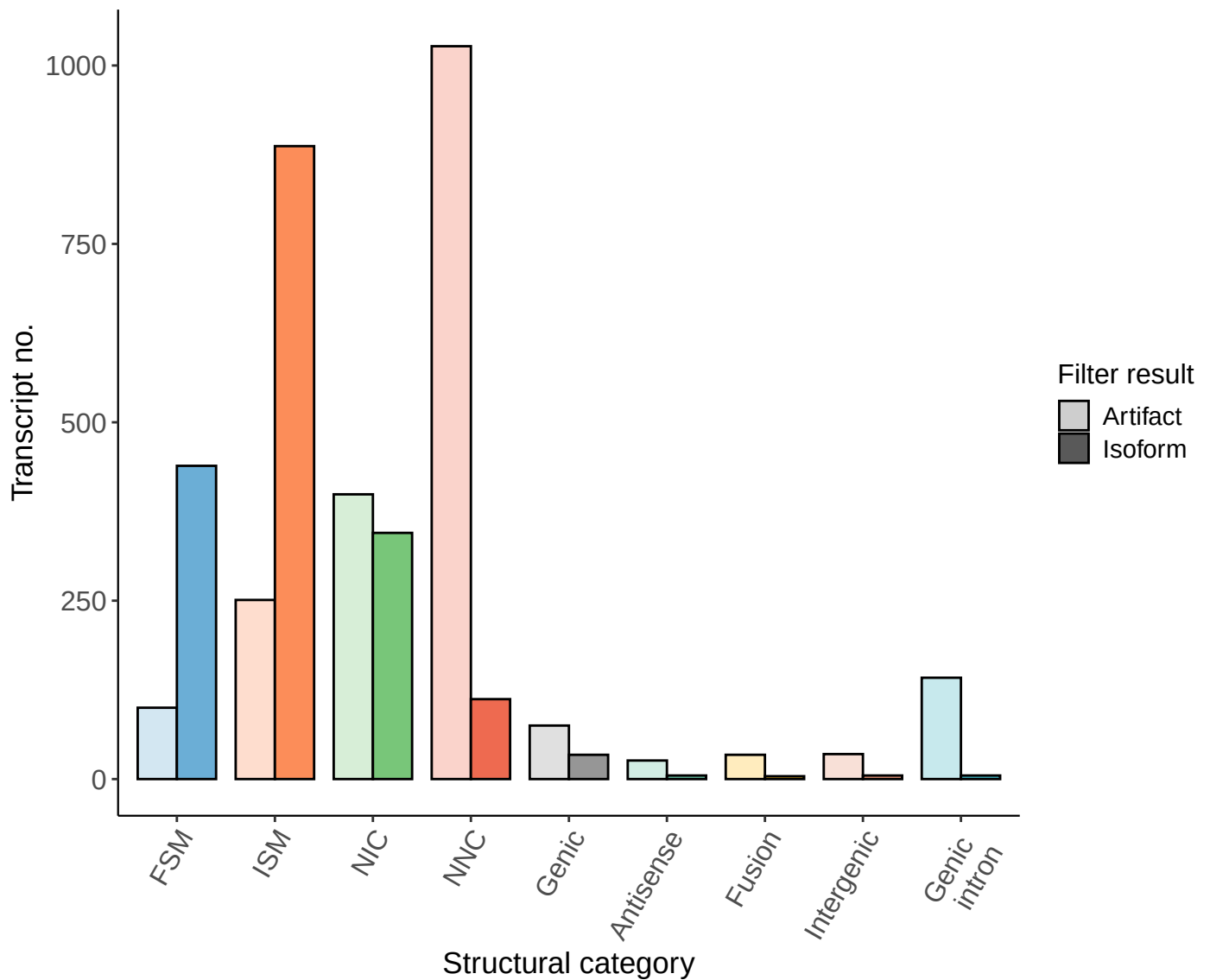
– Isoforms: 1836 (47%)

– Artifacts: 2089 (53%)

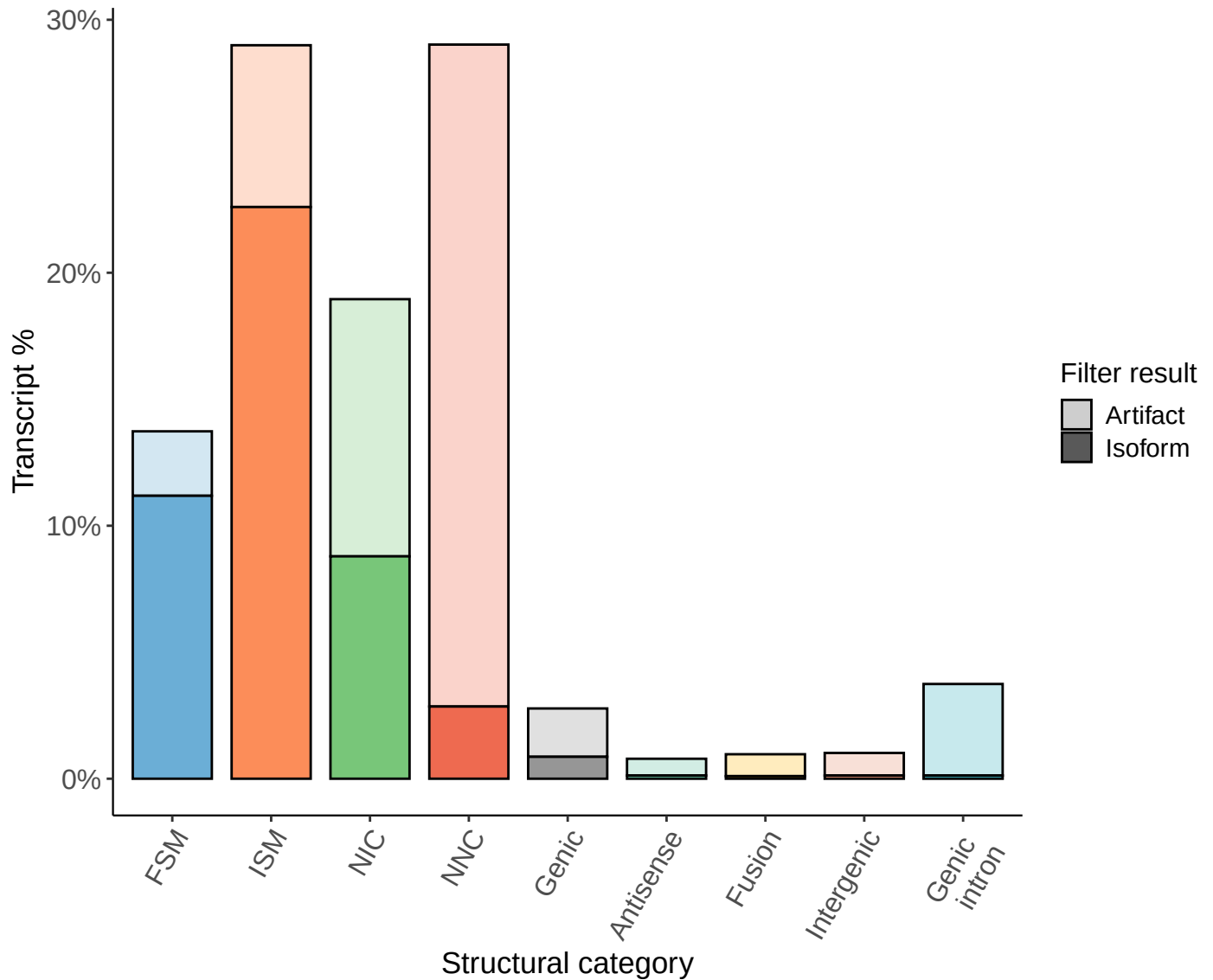
Gene category	Gene no.	No. of genes with artifacts only
Annotated	451	90
Novel	216	201

Structural category	Artifact no.	Isoform no.
FSM	100	439
ISM	251	887
NIC	399	345
NNC	1027	112
Genic	75	34
Antisense	26	5
Fusion	34	4
Intergenic	35	5
Genic intron	142	5

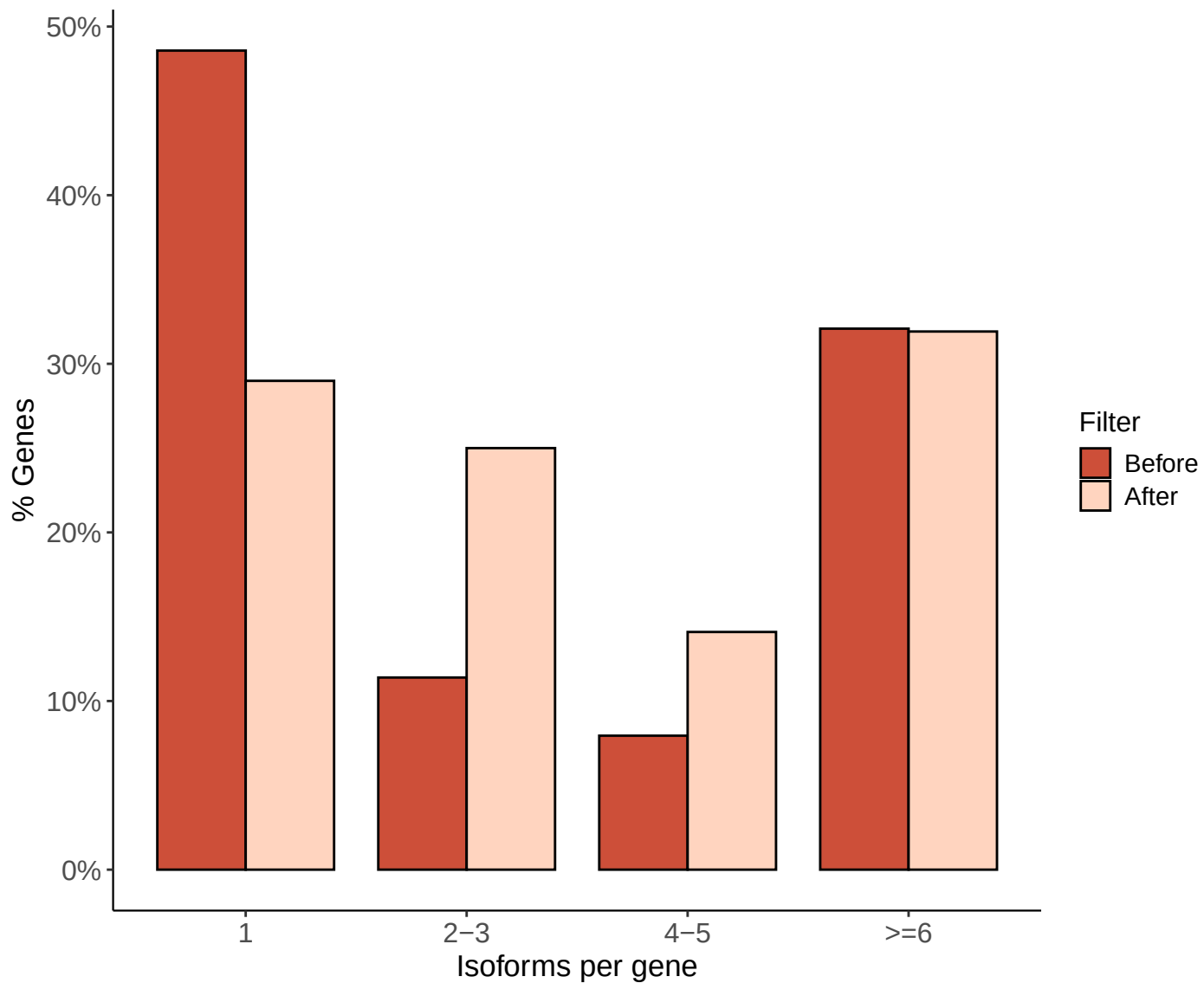
Total isoforms and artifacts by category



% isoforms and artifacts by category

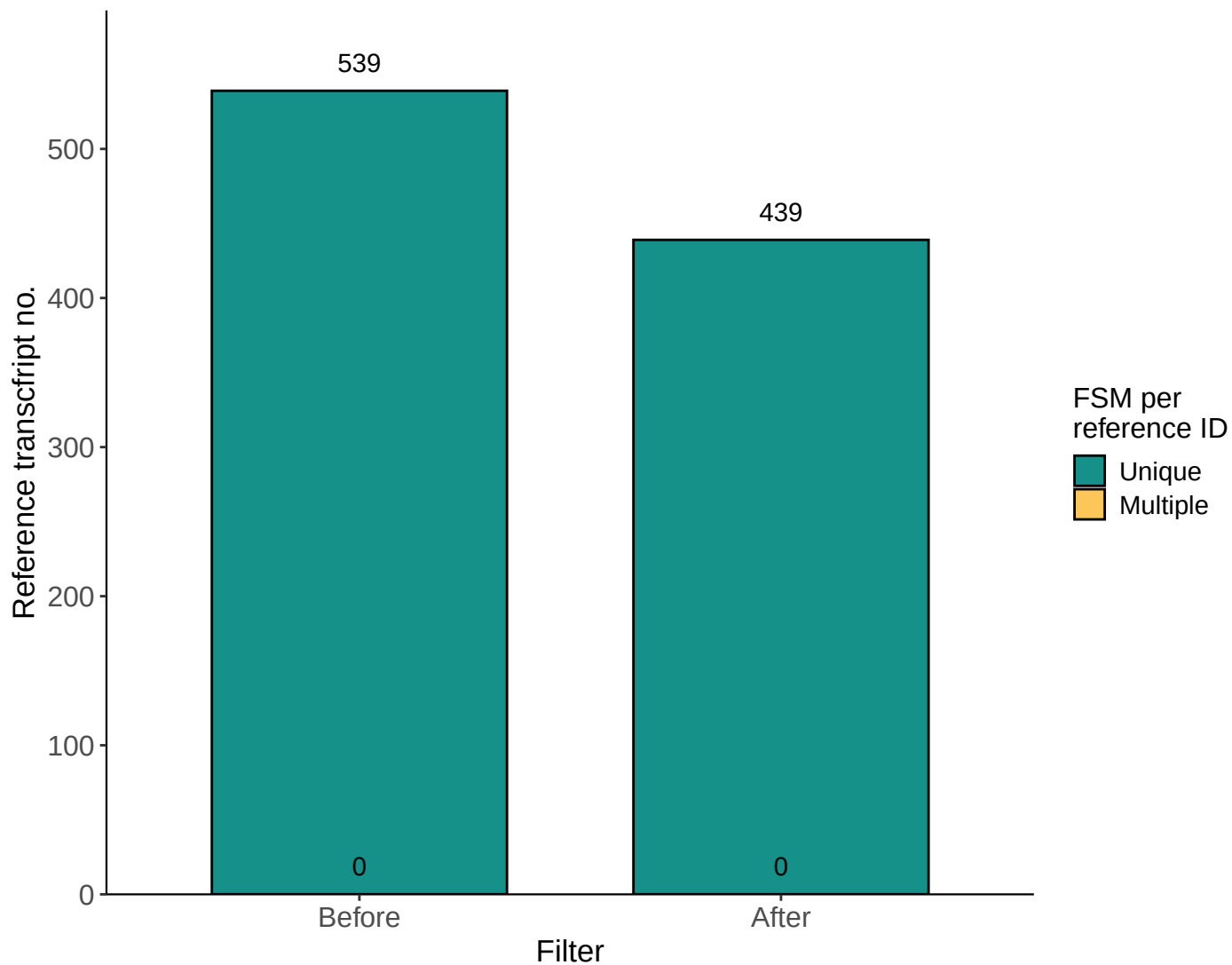


Number of isoforms per gene

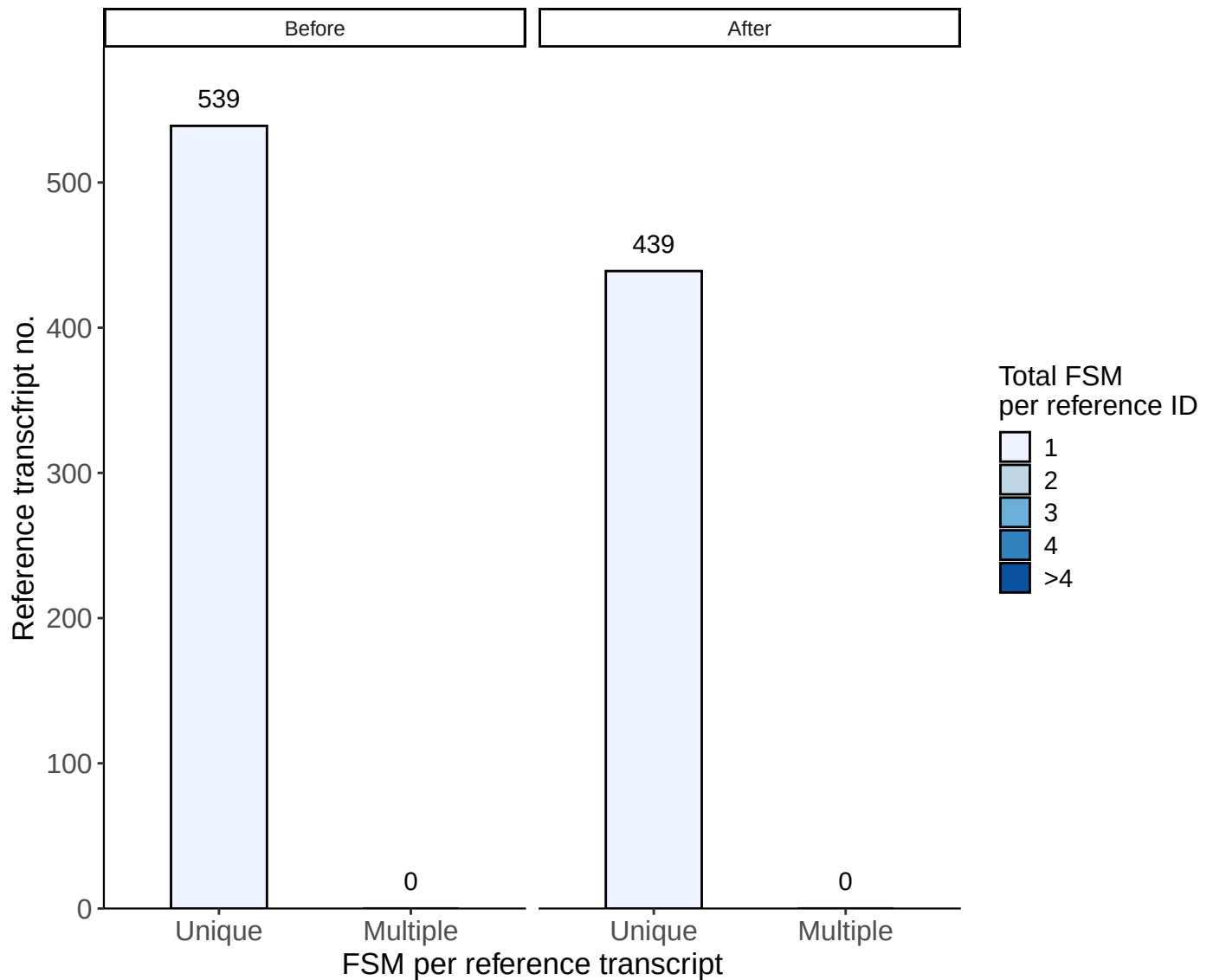


Reference transcript complexity

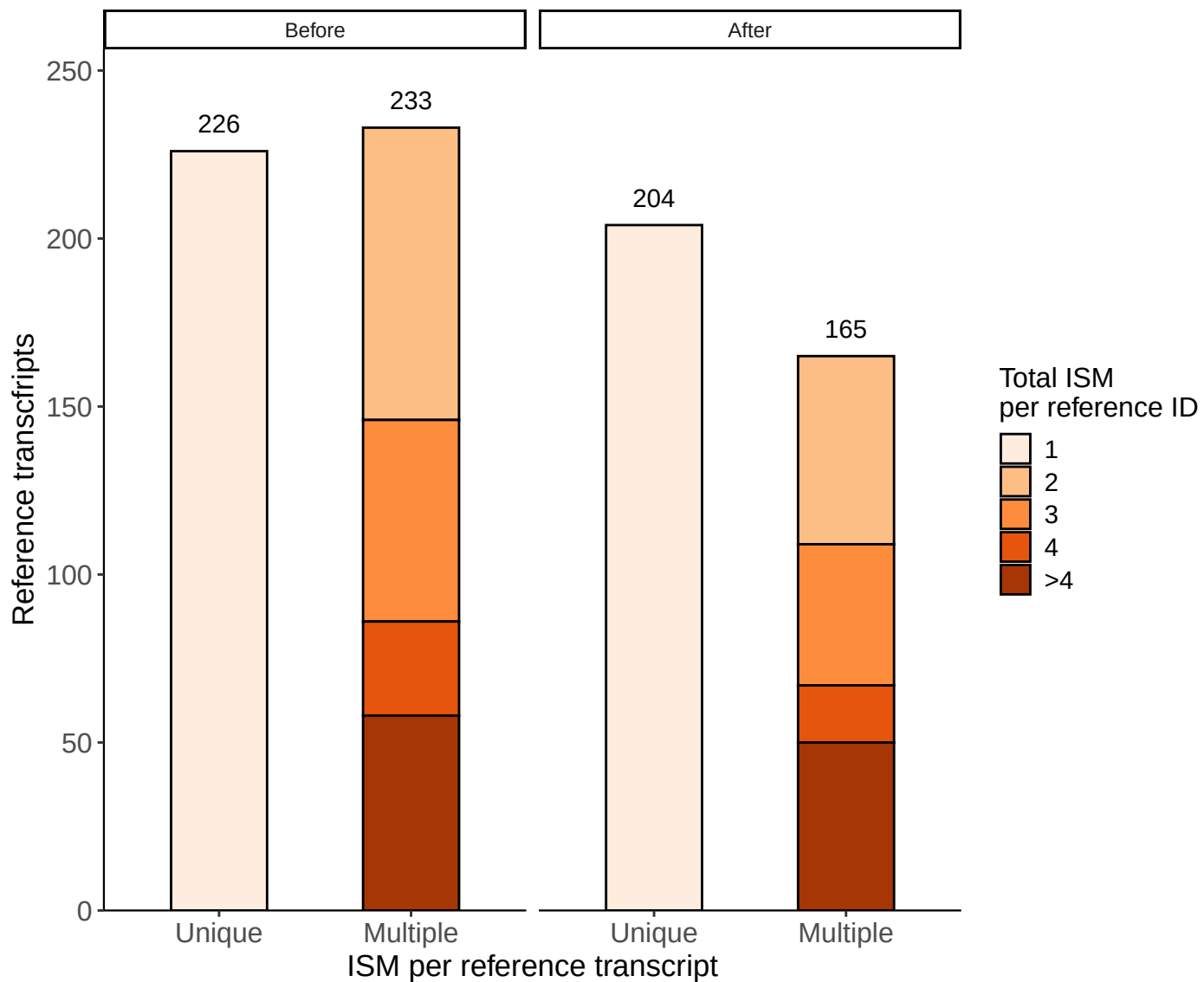
No. of reference transcripts represented by FSM



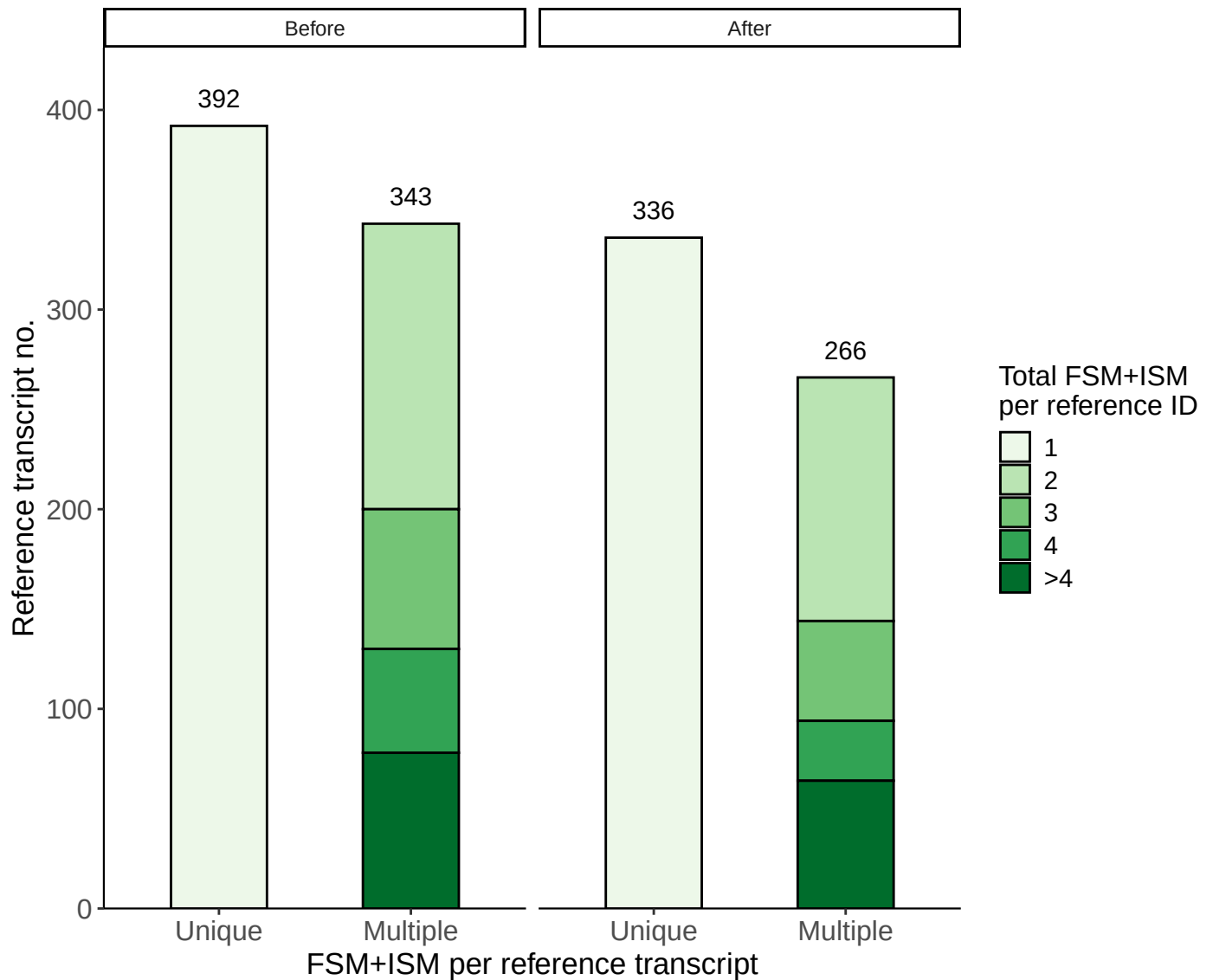
FSM redundancy



ISM redundancy



FSM+ISM redundancy



ML classifier performance report

Classification model performance on test set

Performance metrics

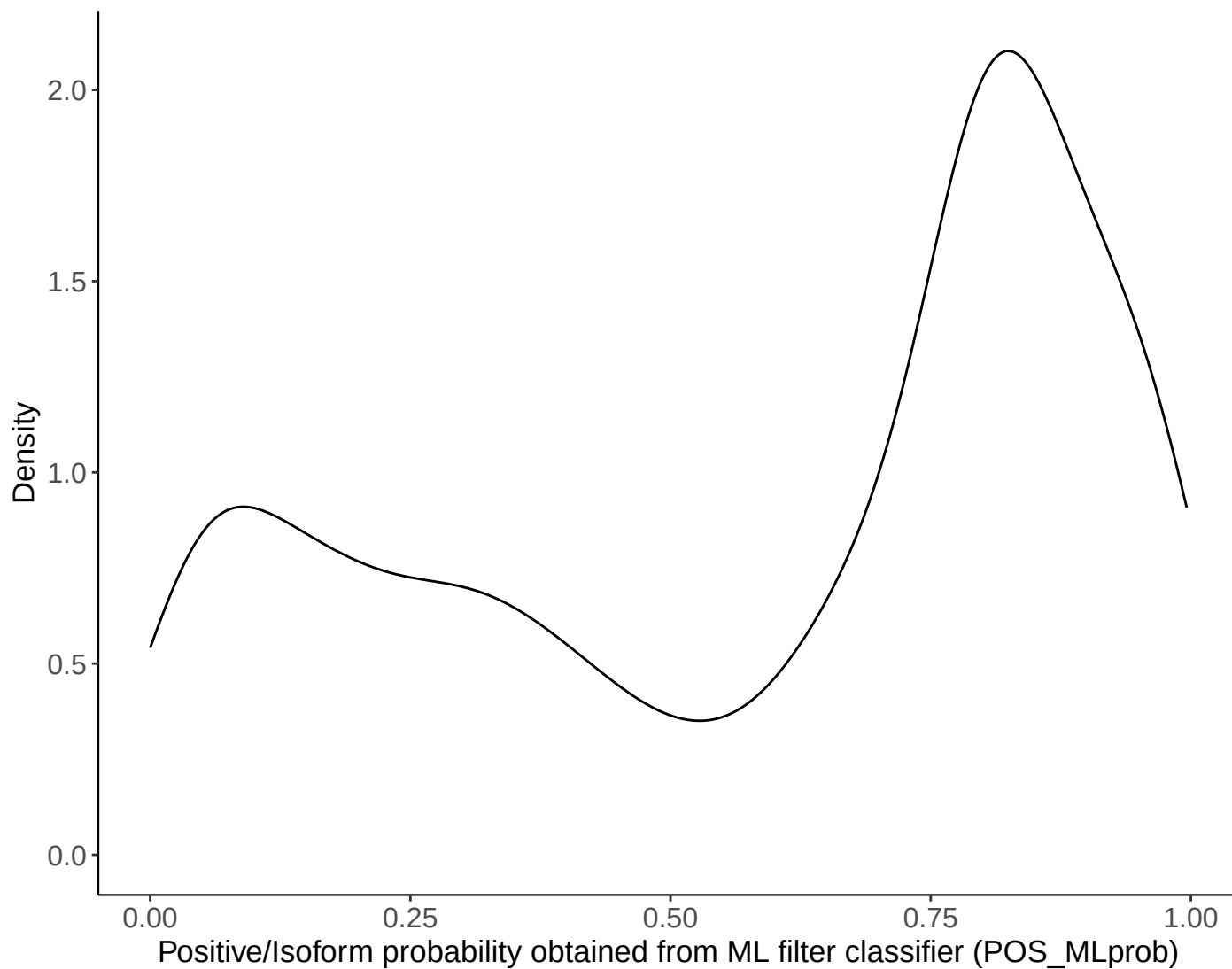
Metric	Value
Accuracy	0.921
Kappa	0.842
AccuracyLower	0.855
AccuracyUpper	0.963
AccuracyNull	0.5
McnemarPValue	0.0455
Sensitivity	0.860
Specificity	0.982
Pos Pred Value	0.98
Neg Pred Value	0.875
Precision	0.98
Recall	0.860
F1	0.916
Prevalence	0.5
Detection Rate	0.430
Detection Prevalence	0.439
Balanced Accuracy	0.921

Confusion matrix

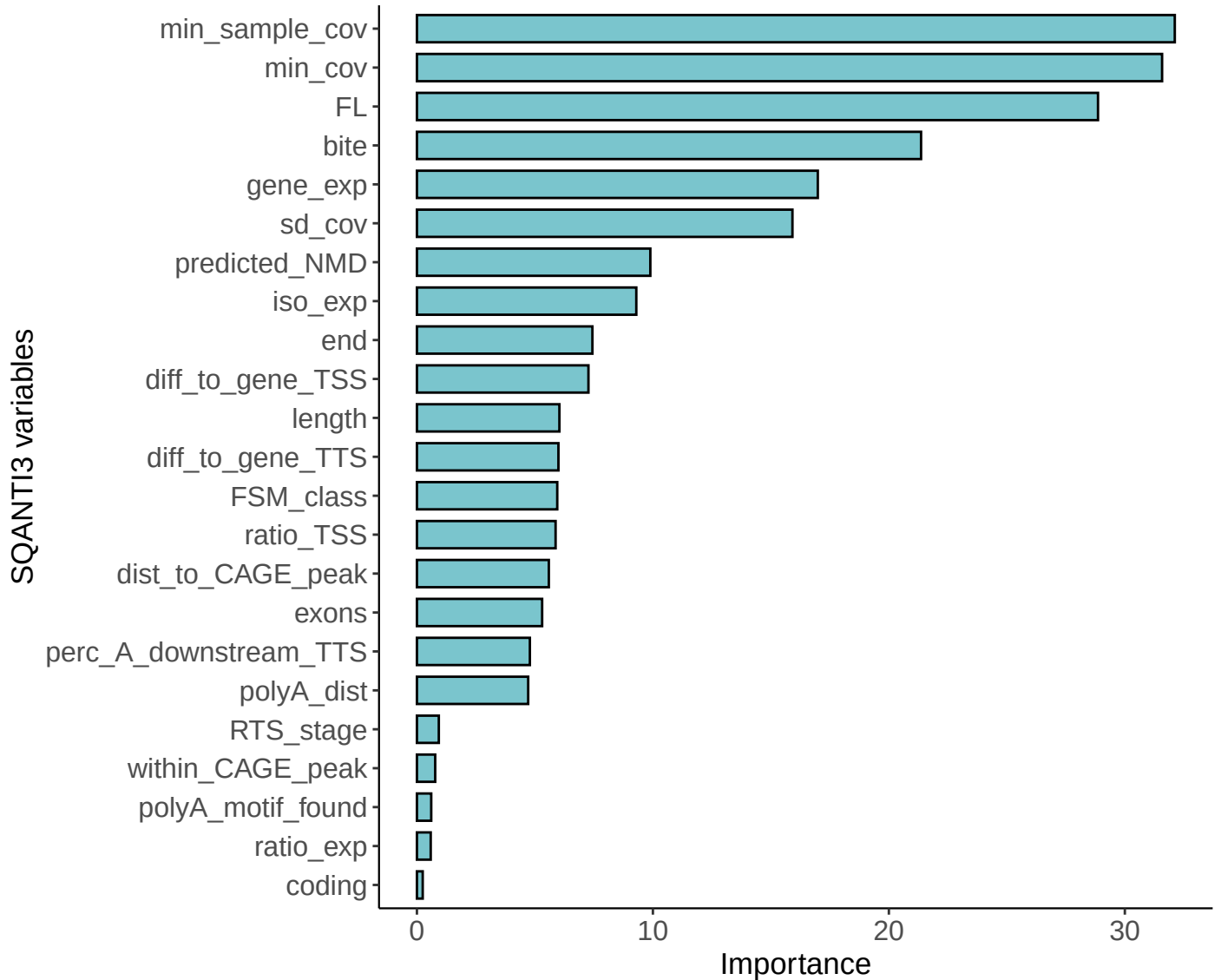
Prediction	Reference	Freq
POS	POS	49
NEG	POS	8
POS	NEG	1
NEG	NEG	56

AccuracyPValue	3.407009e-22
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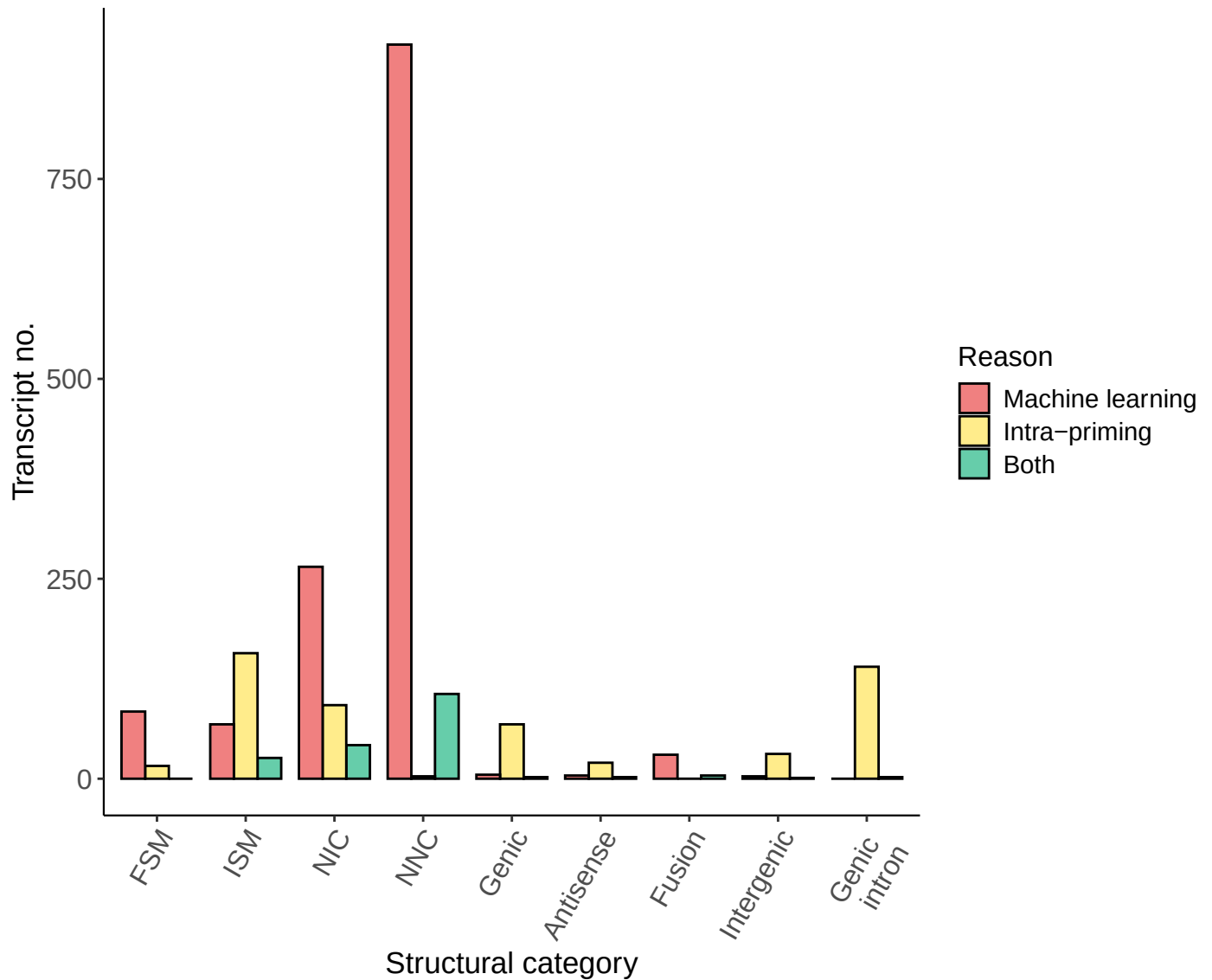
Transcripts classified: 3338



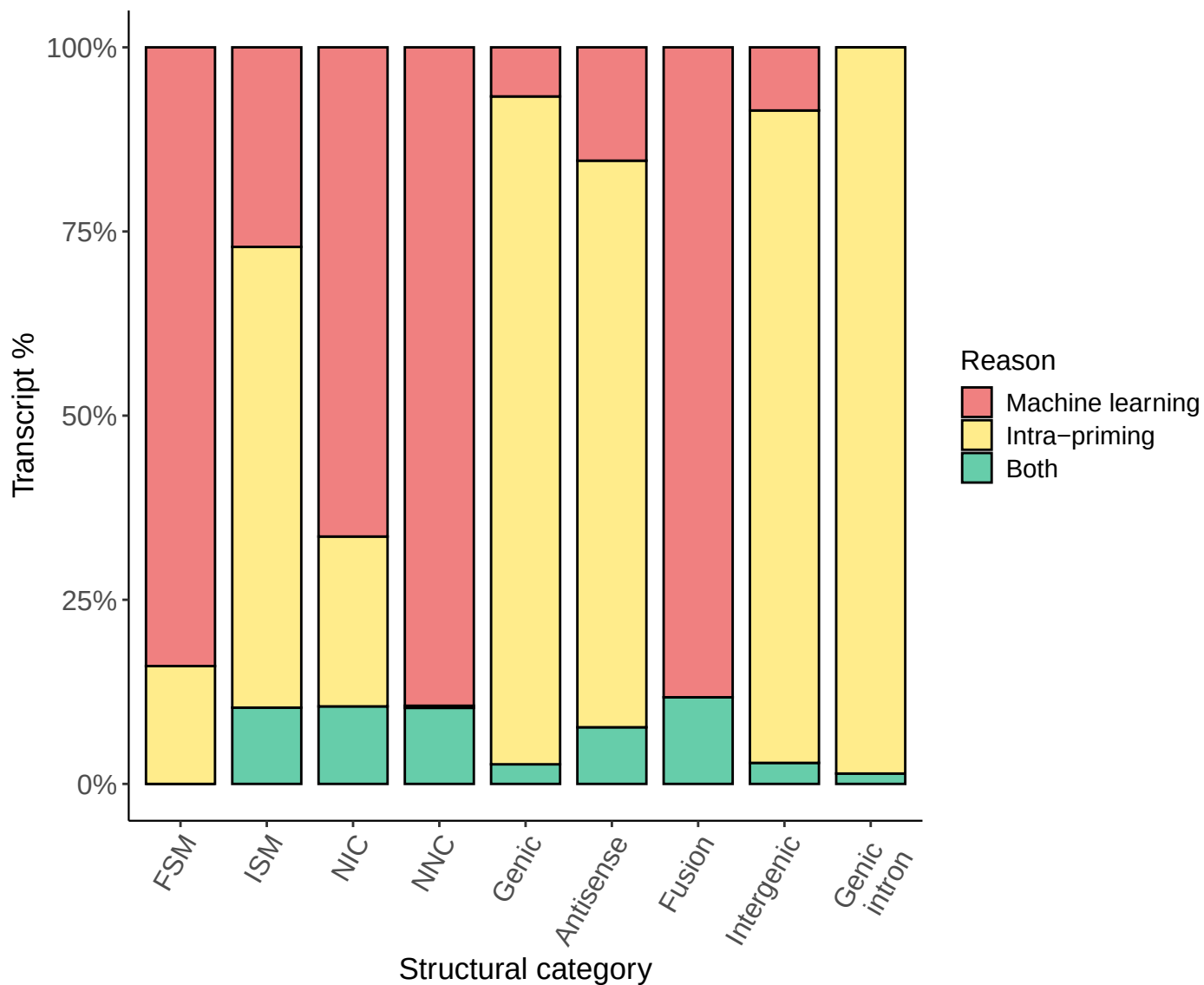
Variable importance in Random Forest classifier



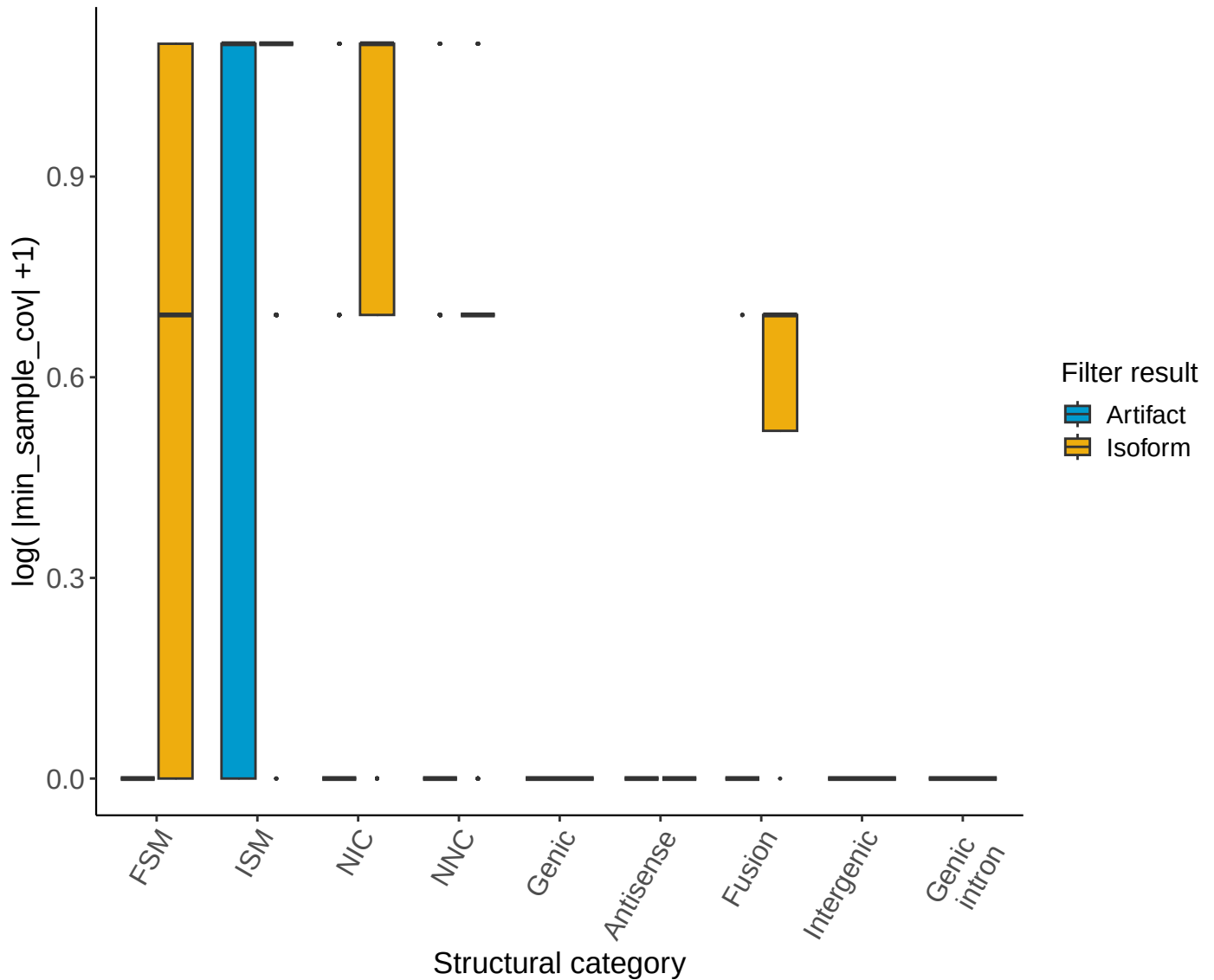
Reason to flag transcripts as artifacts, by category



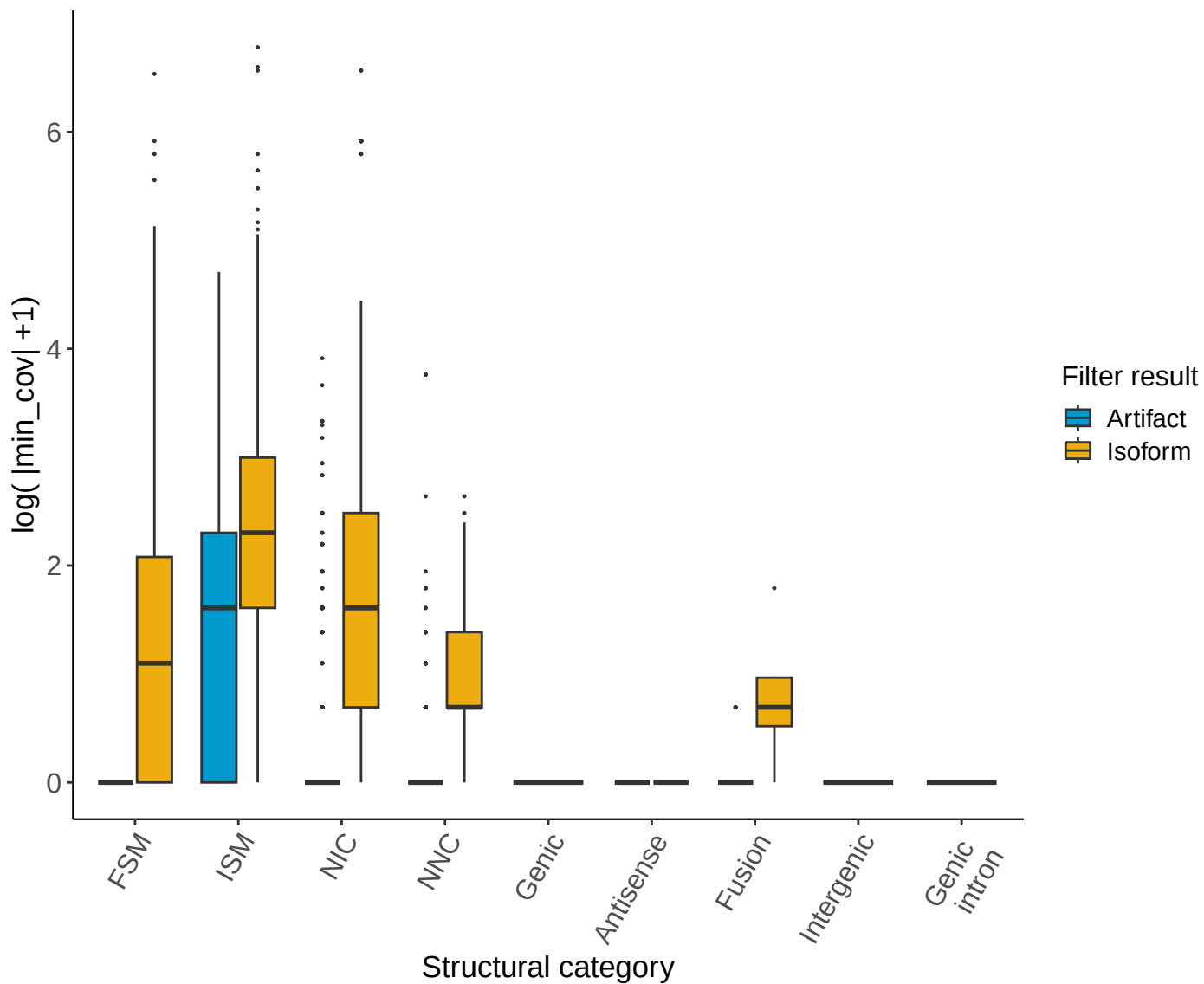
Reason to flag transcripts as artifacts, by category



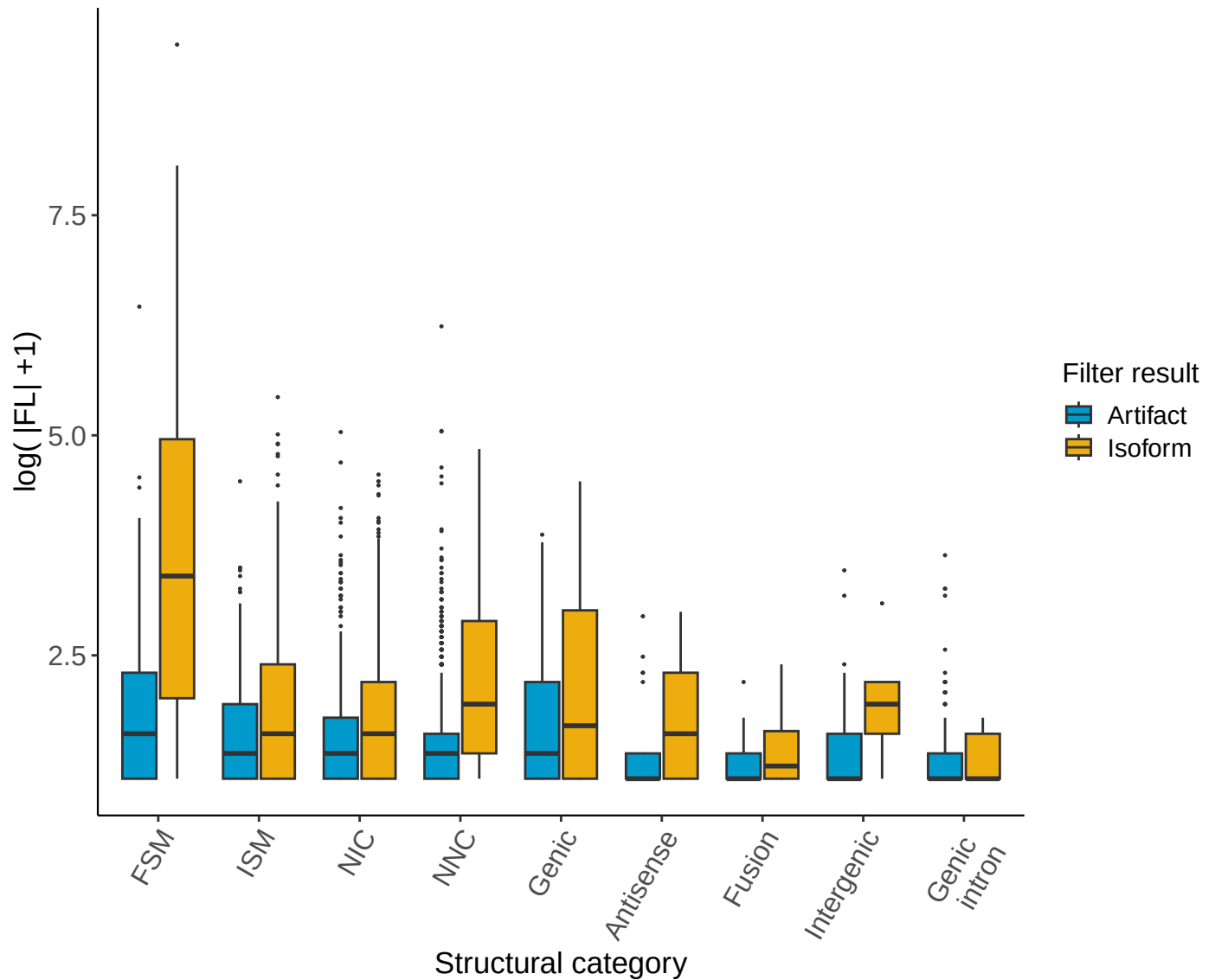
min_sample_cov – ML importance: 32.12



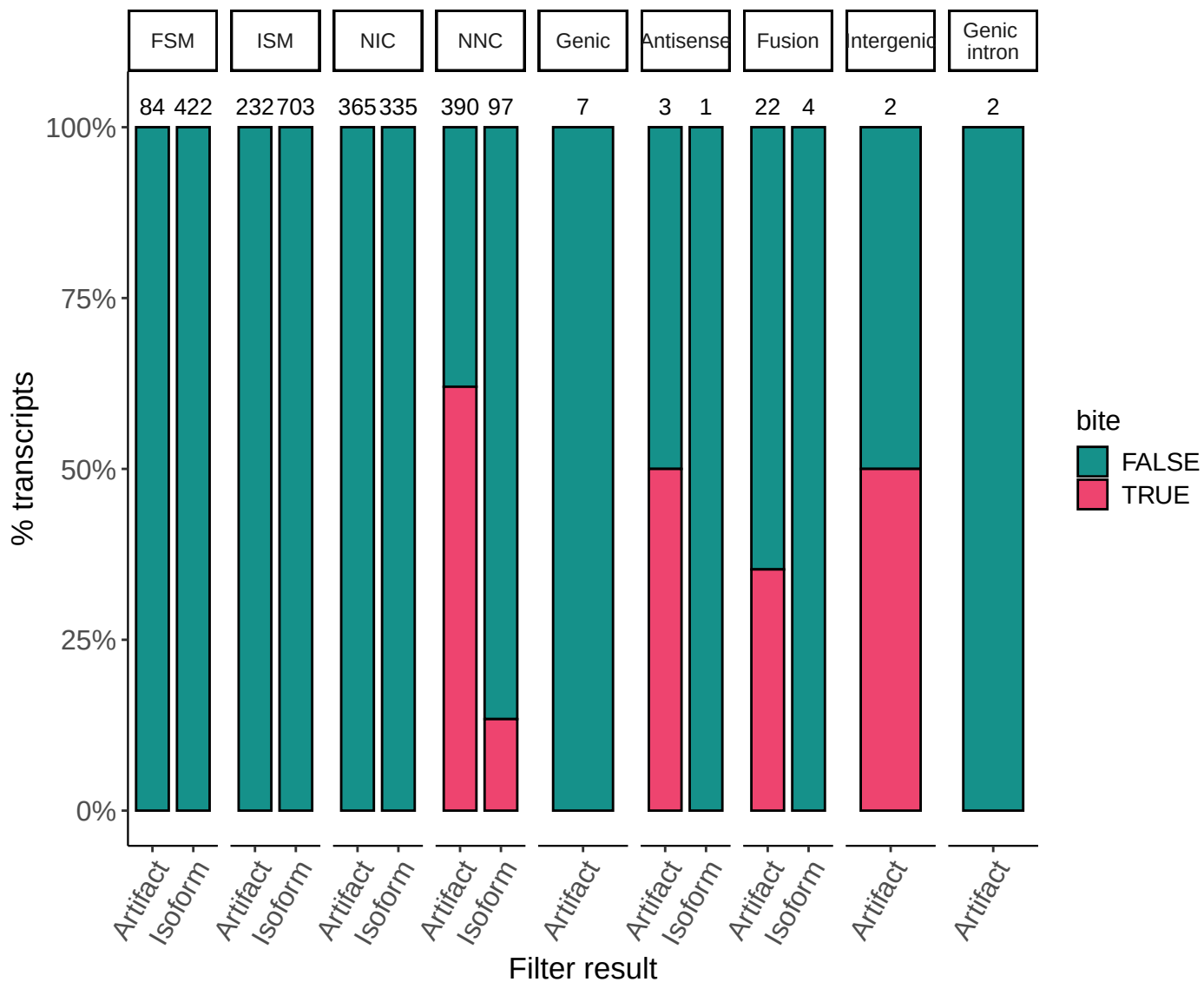
min_cov – ML importance: 31.58



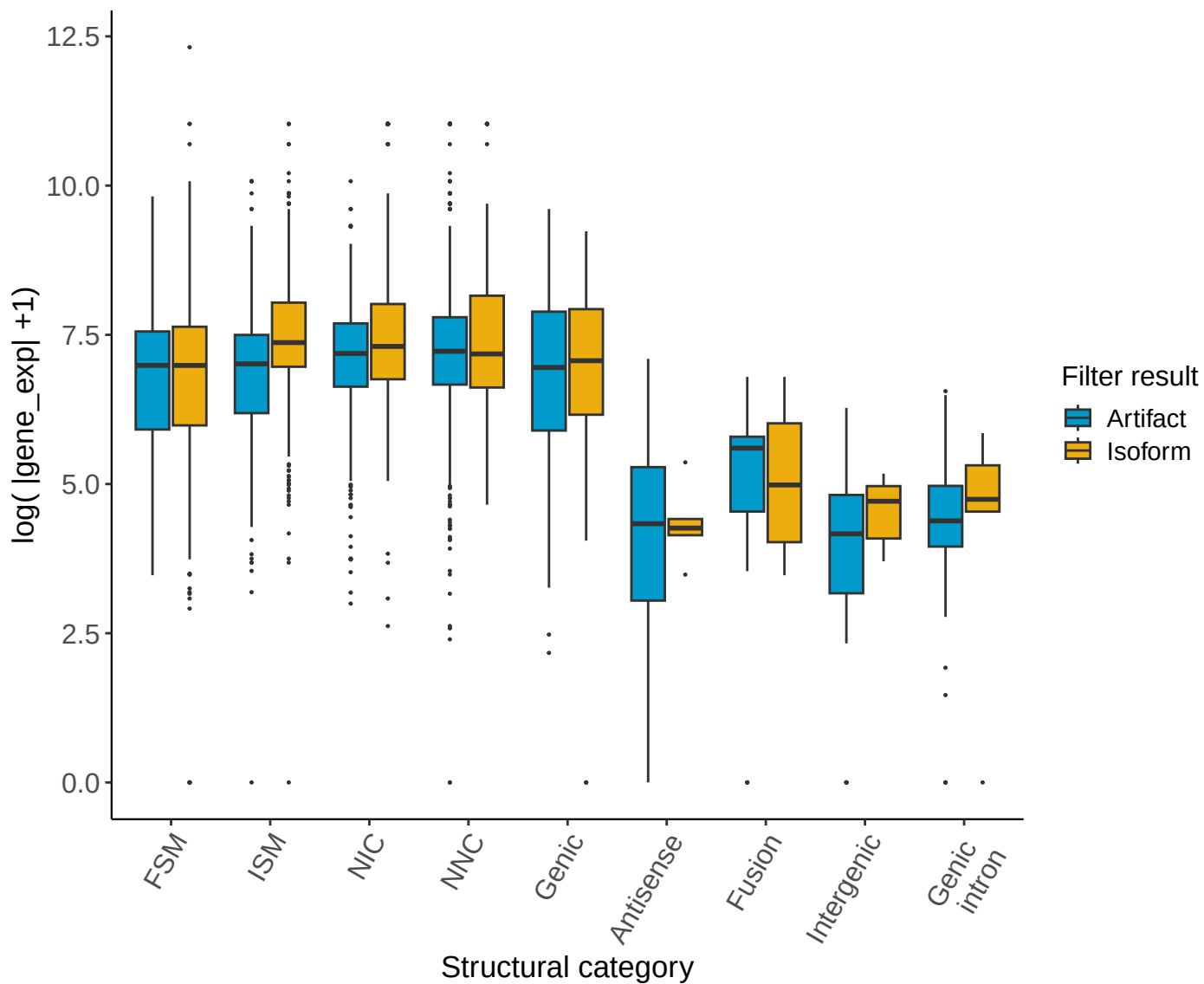
FL – ML importance: 28.86



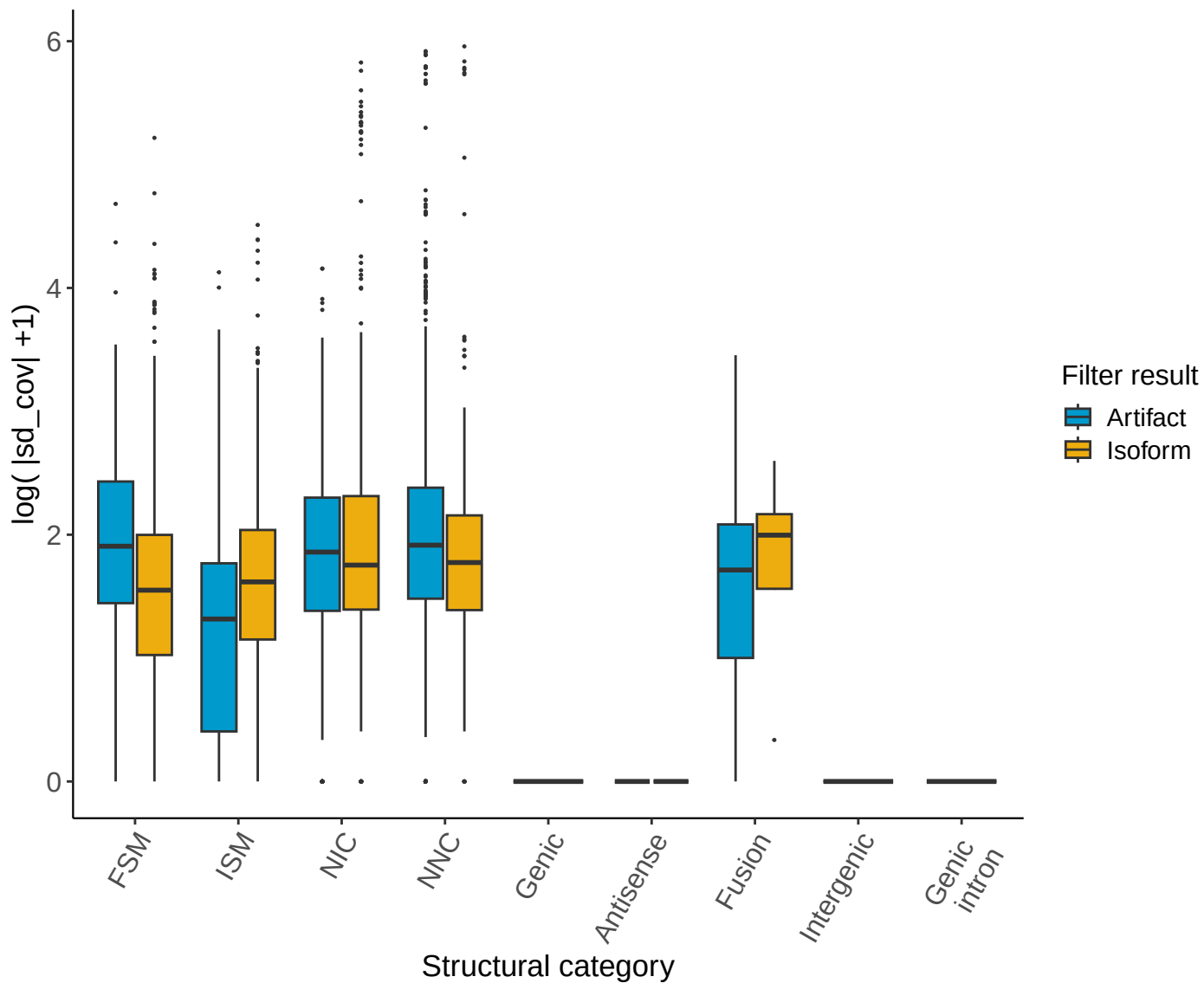
bite – ML importance: 21.36



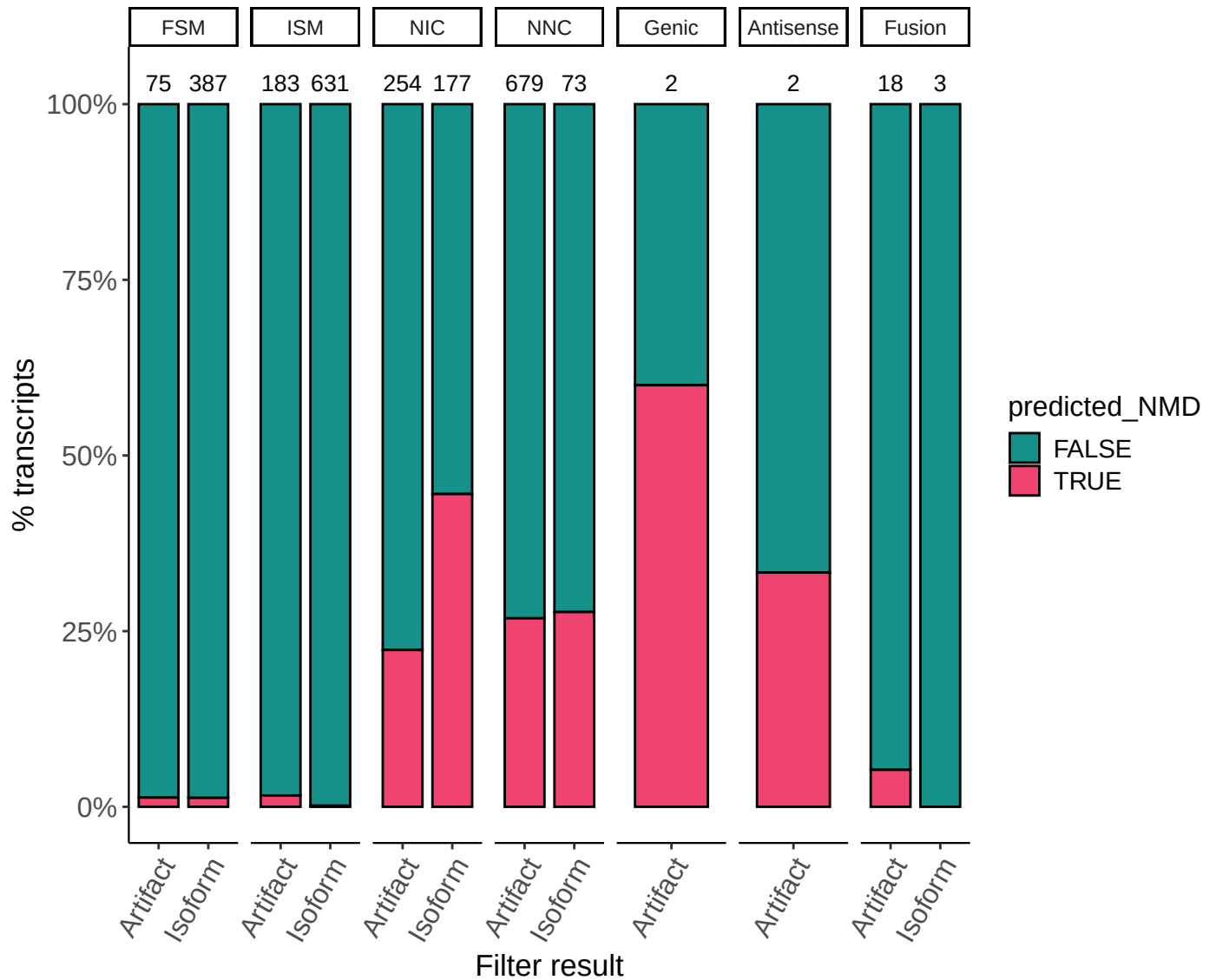
gene_exp – ML importance: 16.99



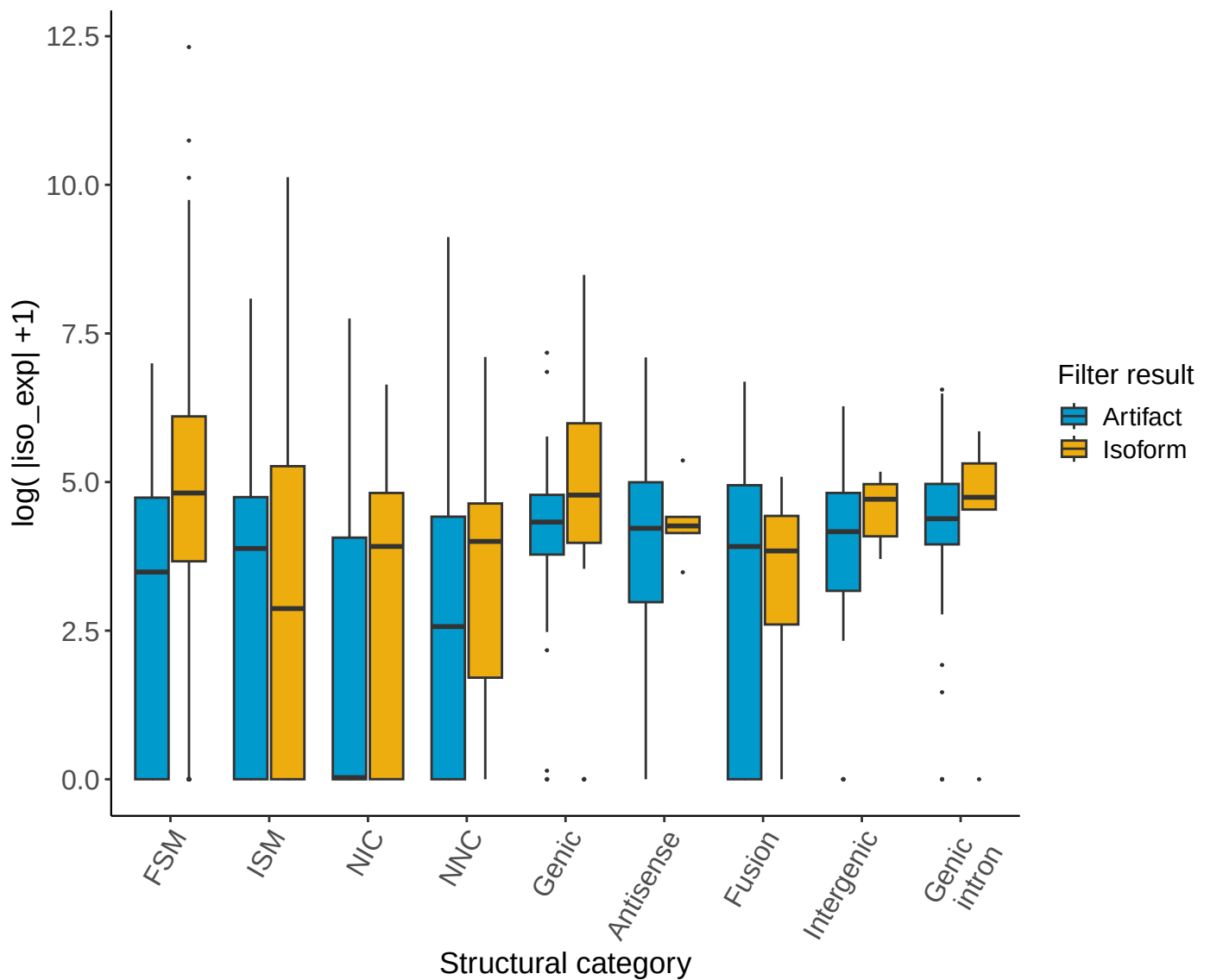
sd_cov – ML importance: 15.91



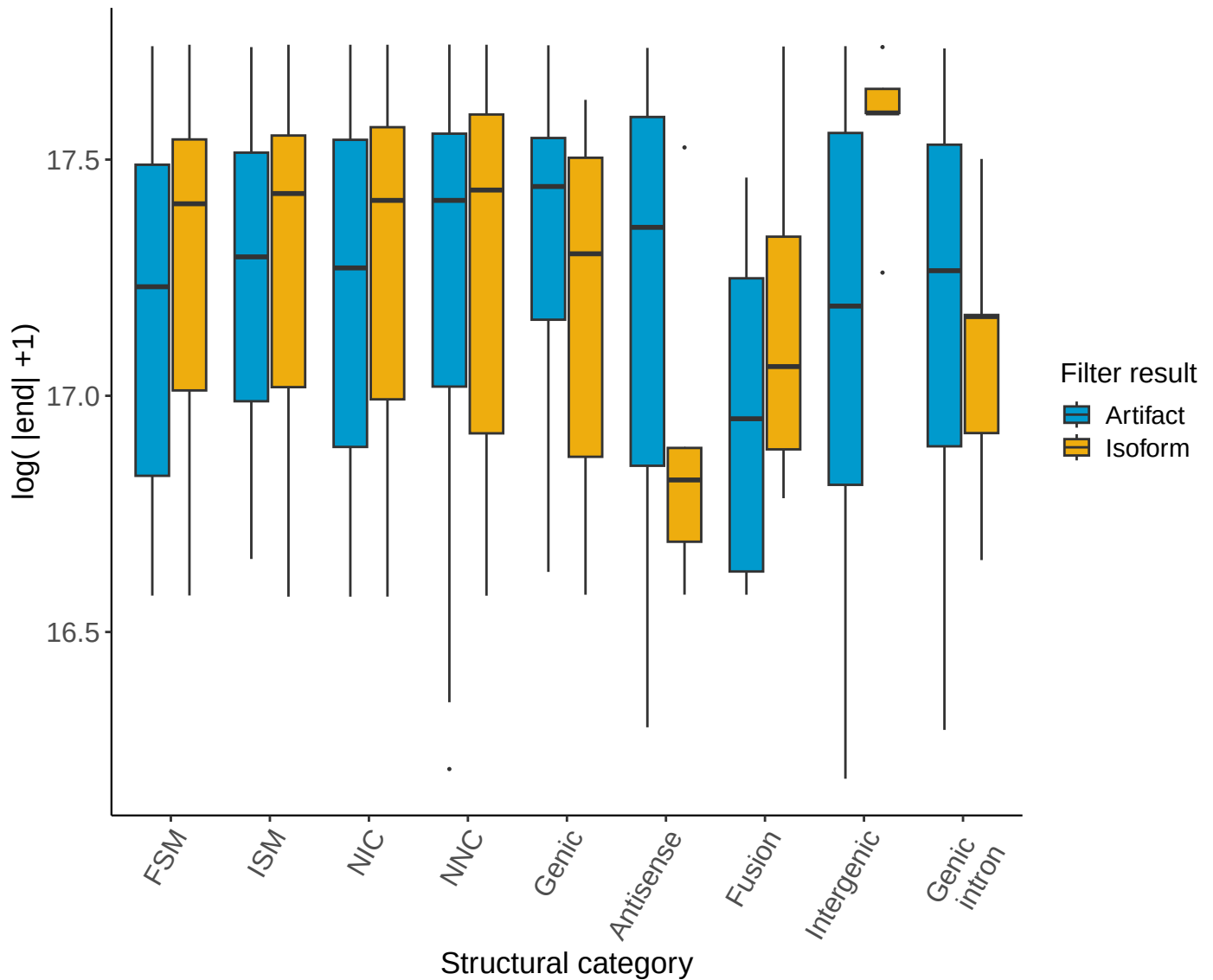
predicted_NMD – ML importance: 9.89



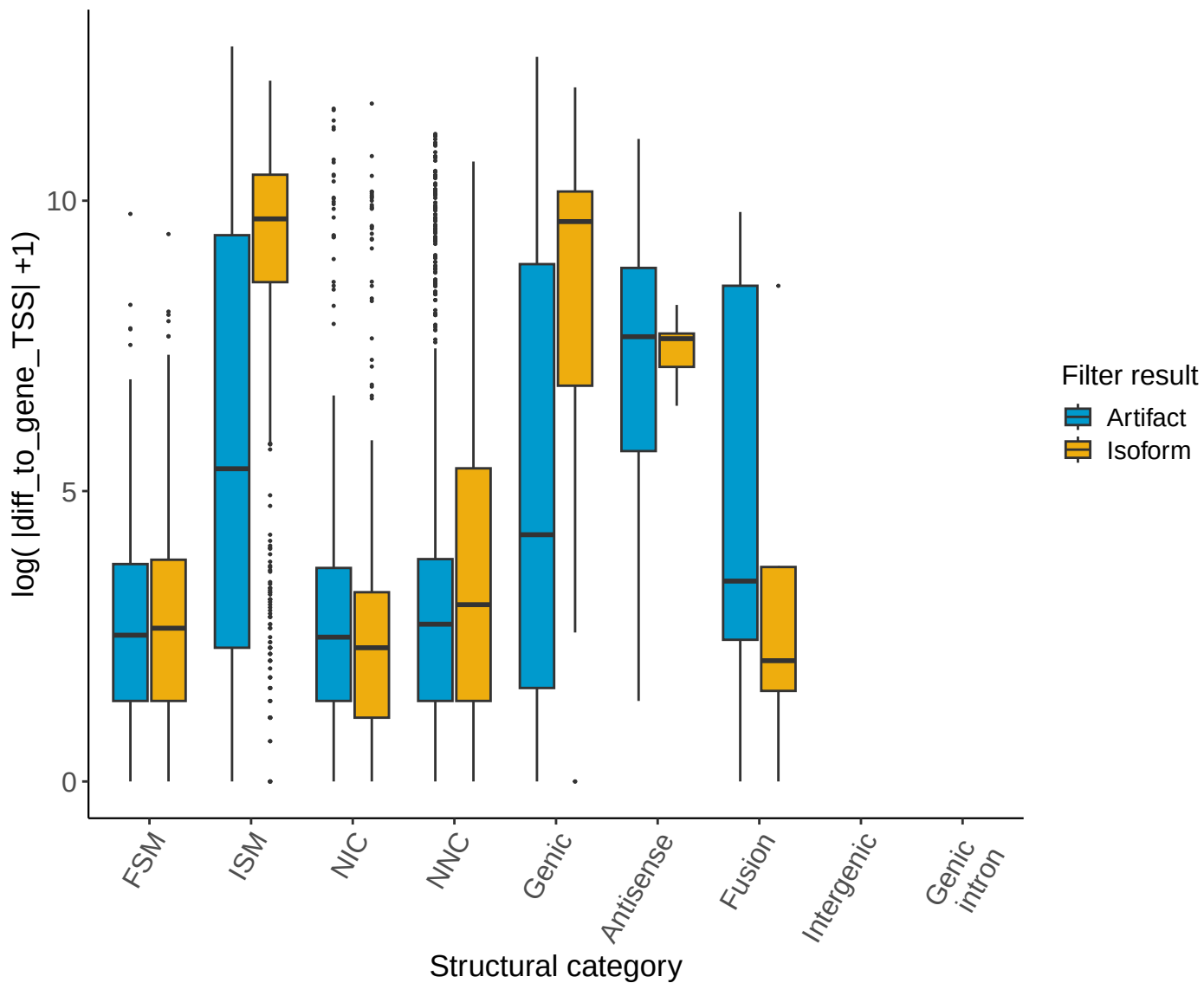
iso_exp – ML importance: 9.3



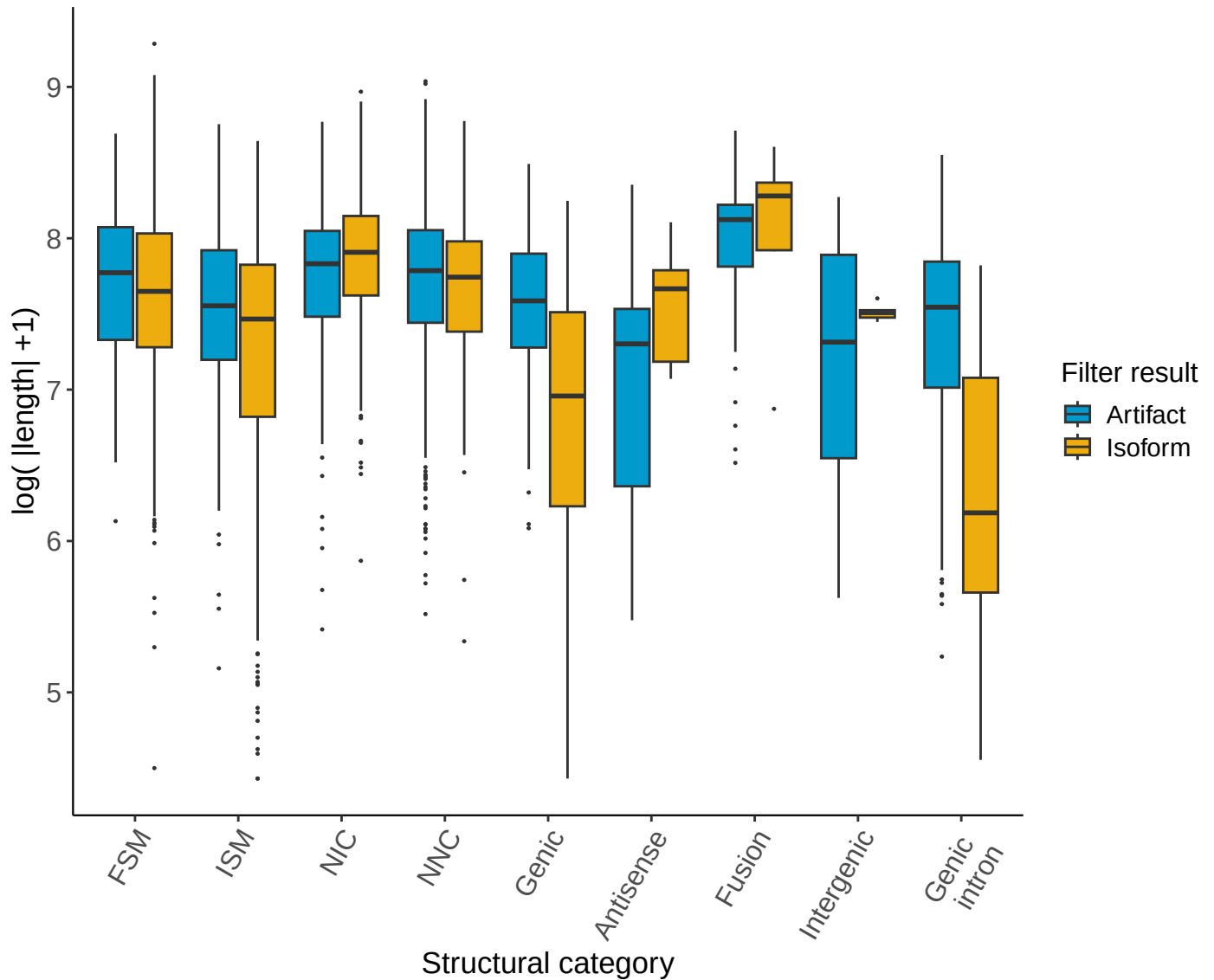
end – ML importance: 7.44



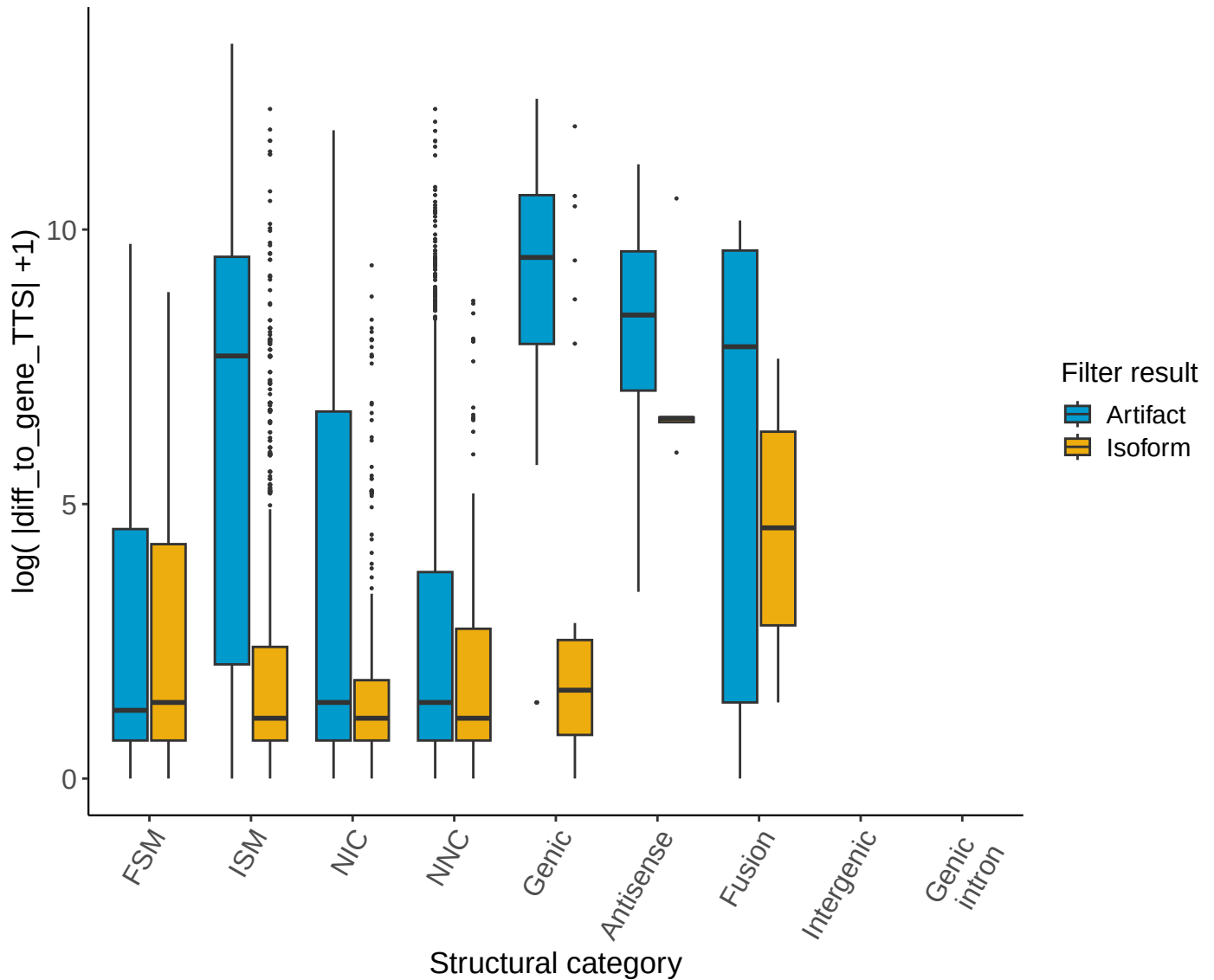
diff_to_gene_TSS – ML importance: 7.27



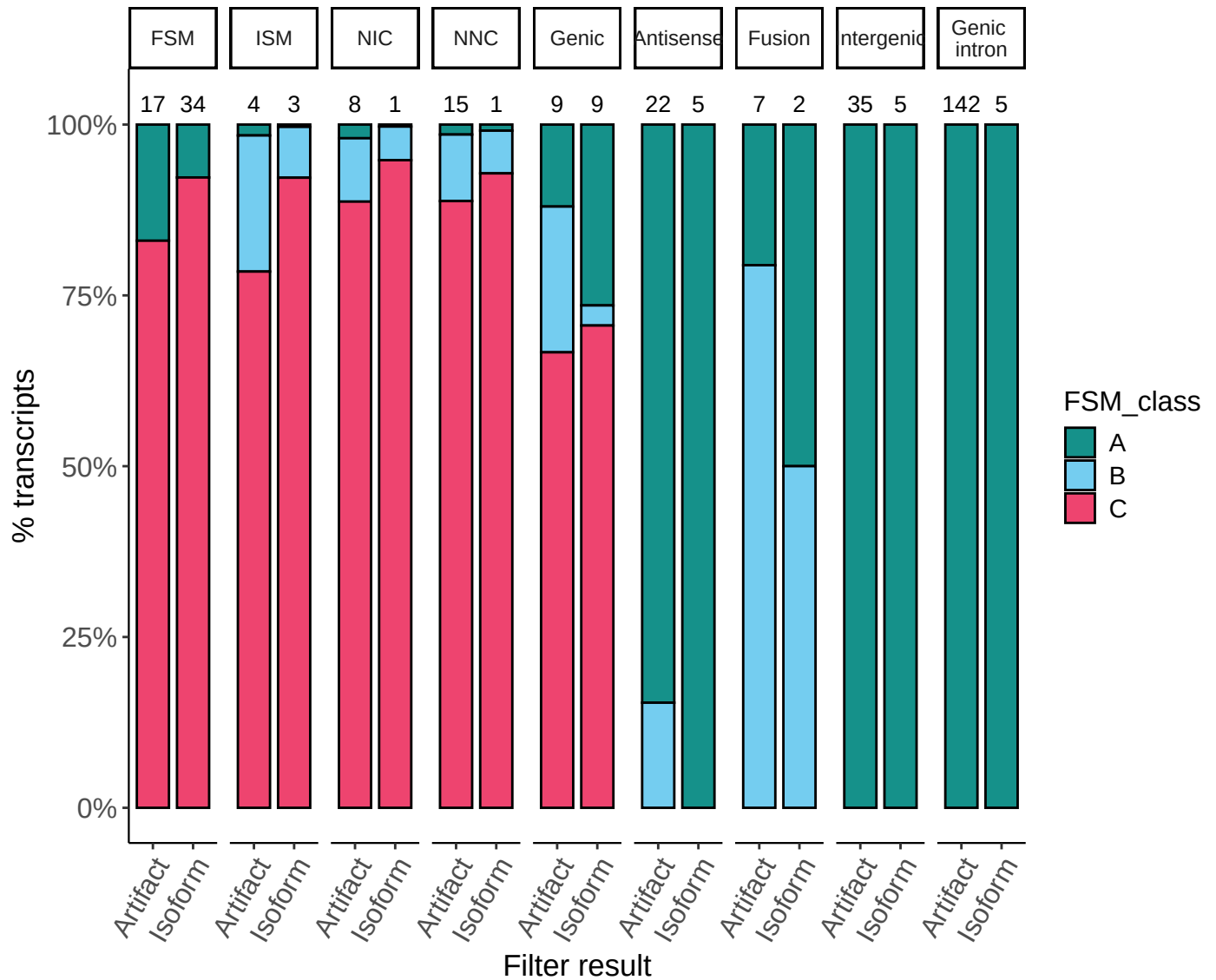
length – ML importance: 6.04



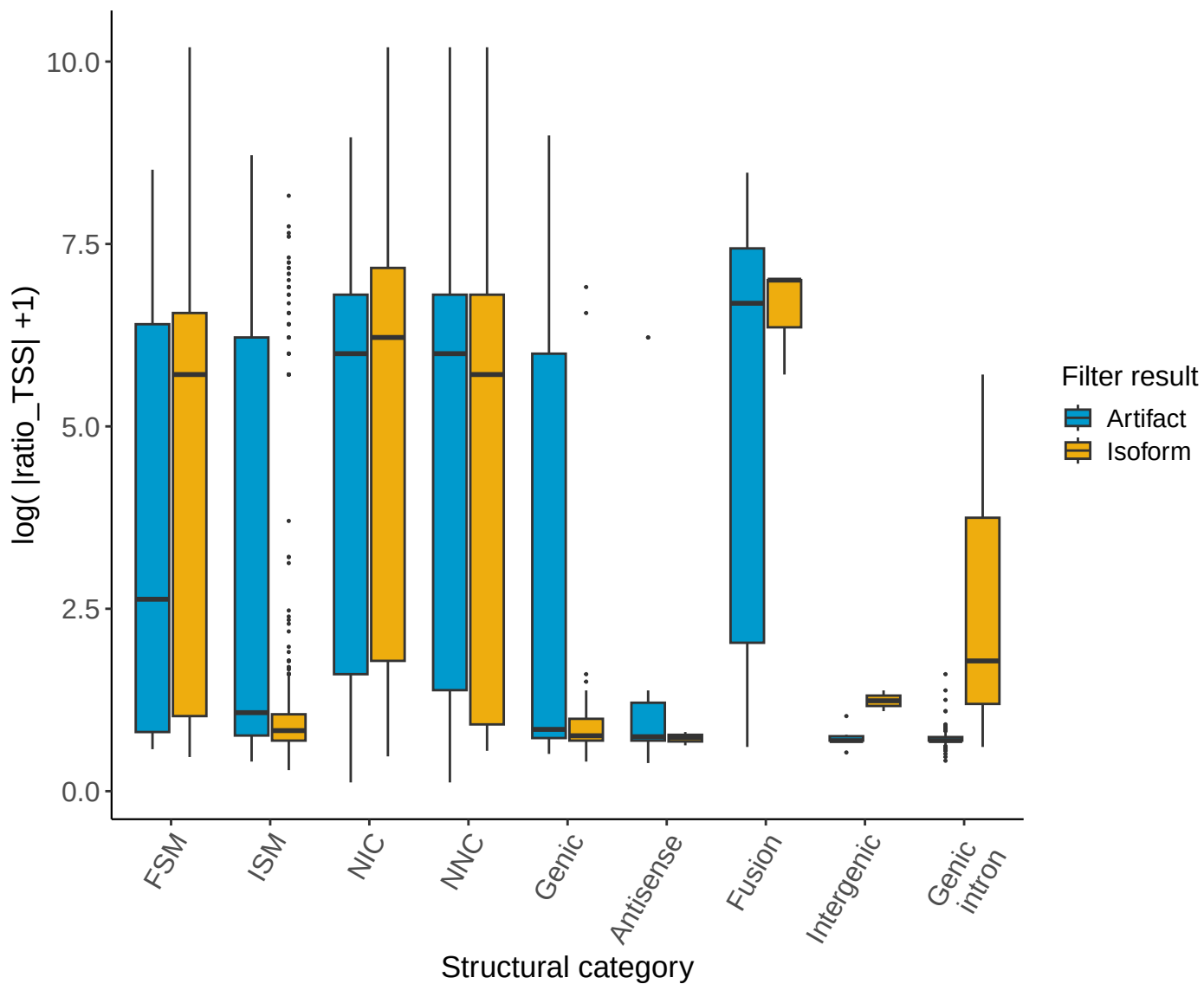
diff_to_gene_TTS – ML importance: 6



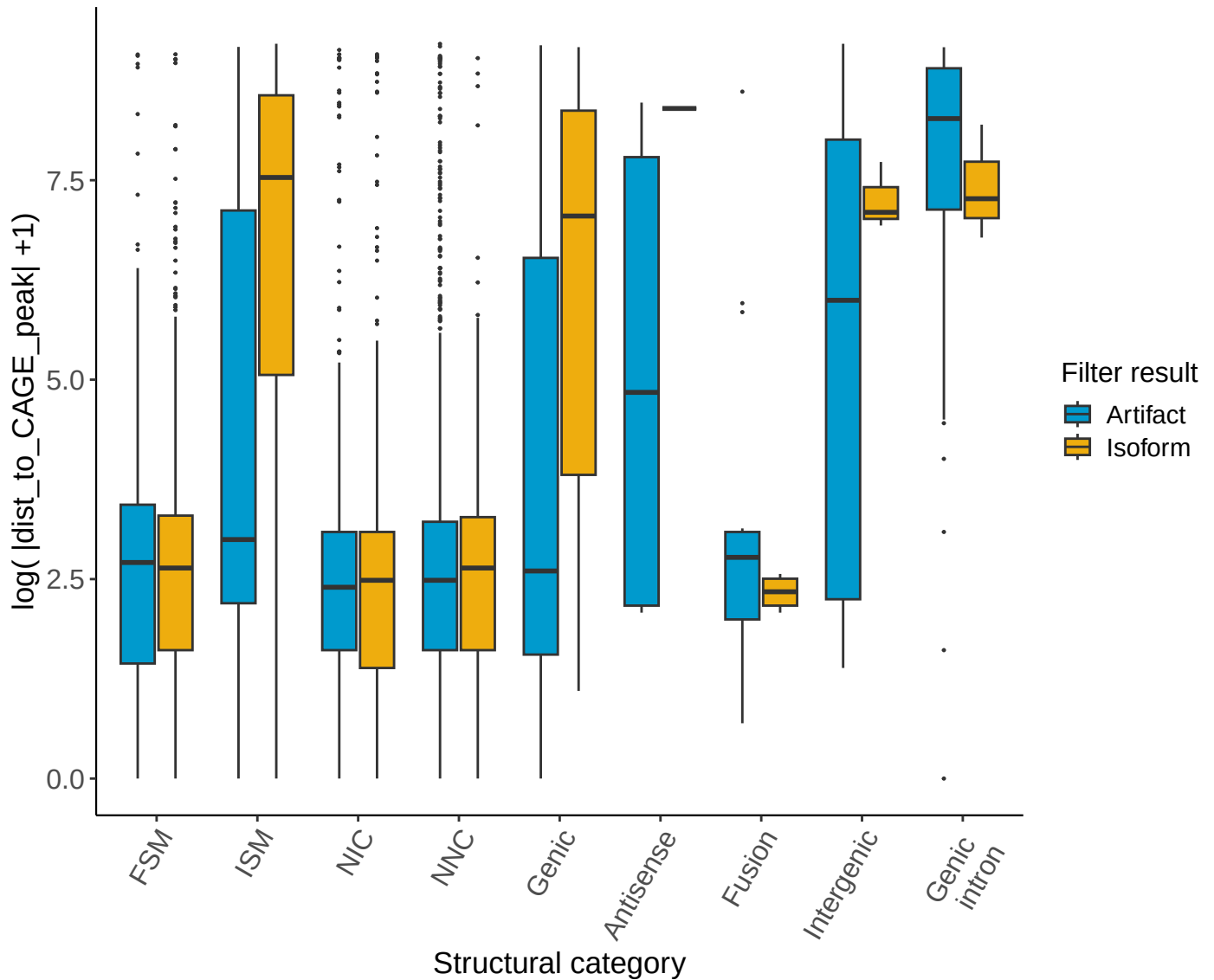
FSM_class – ML importance: 5.95



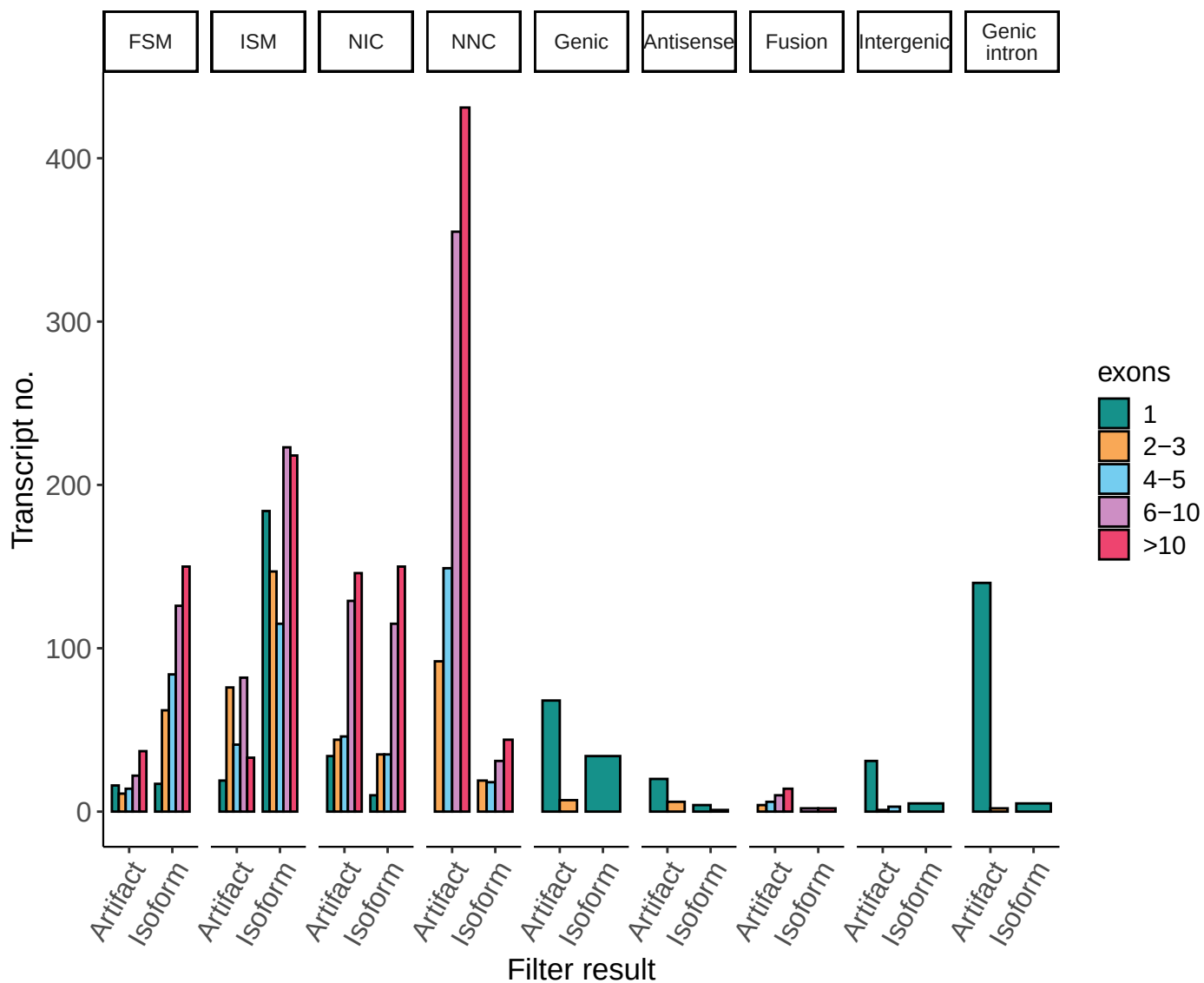
ratio_TSS – ML importance: 5.88



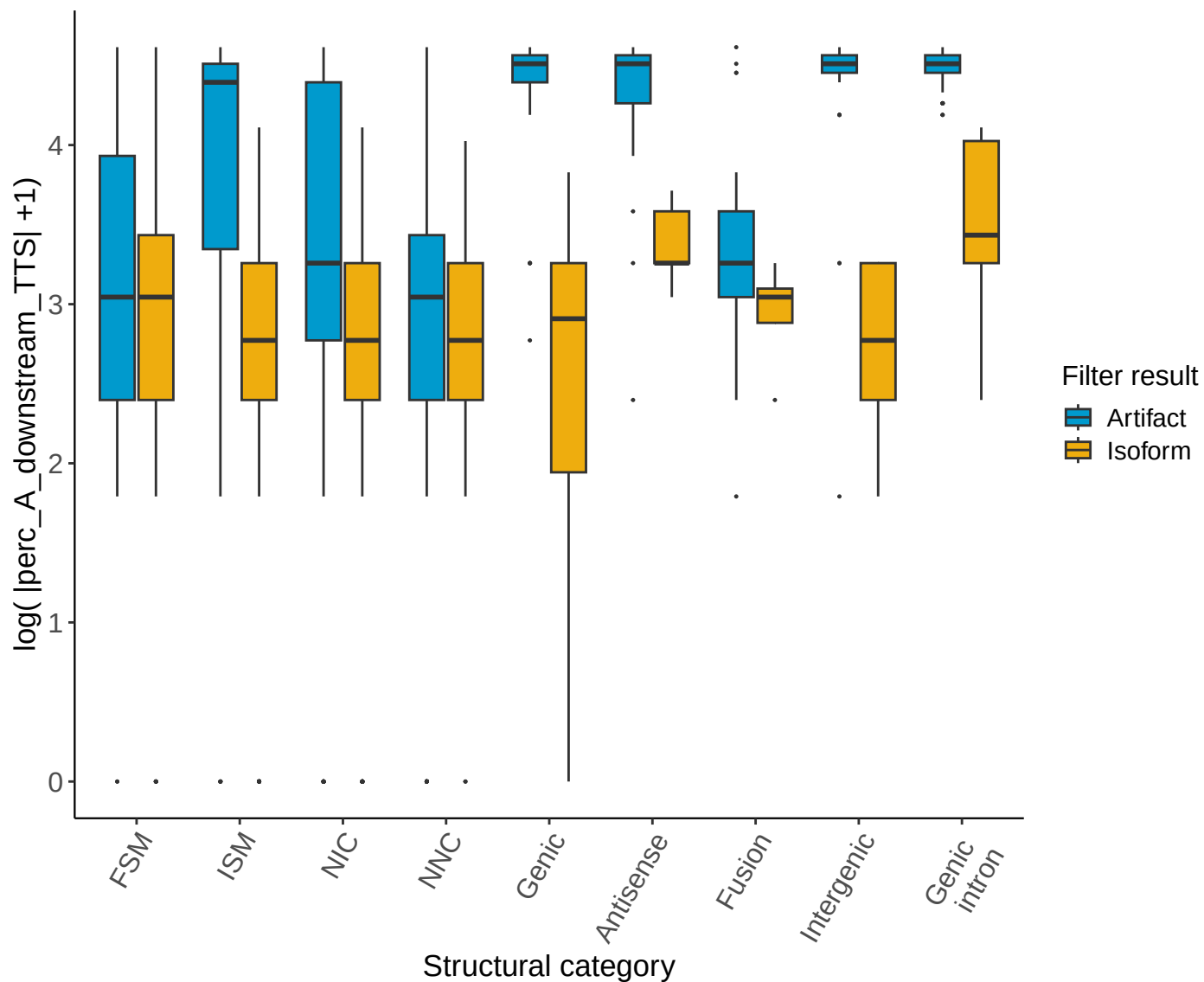
dist_to_CAGE_peak – ML importance: 5.59



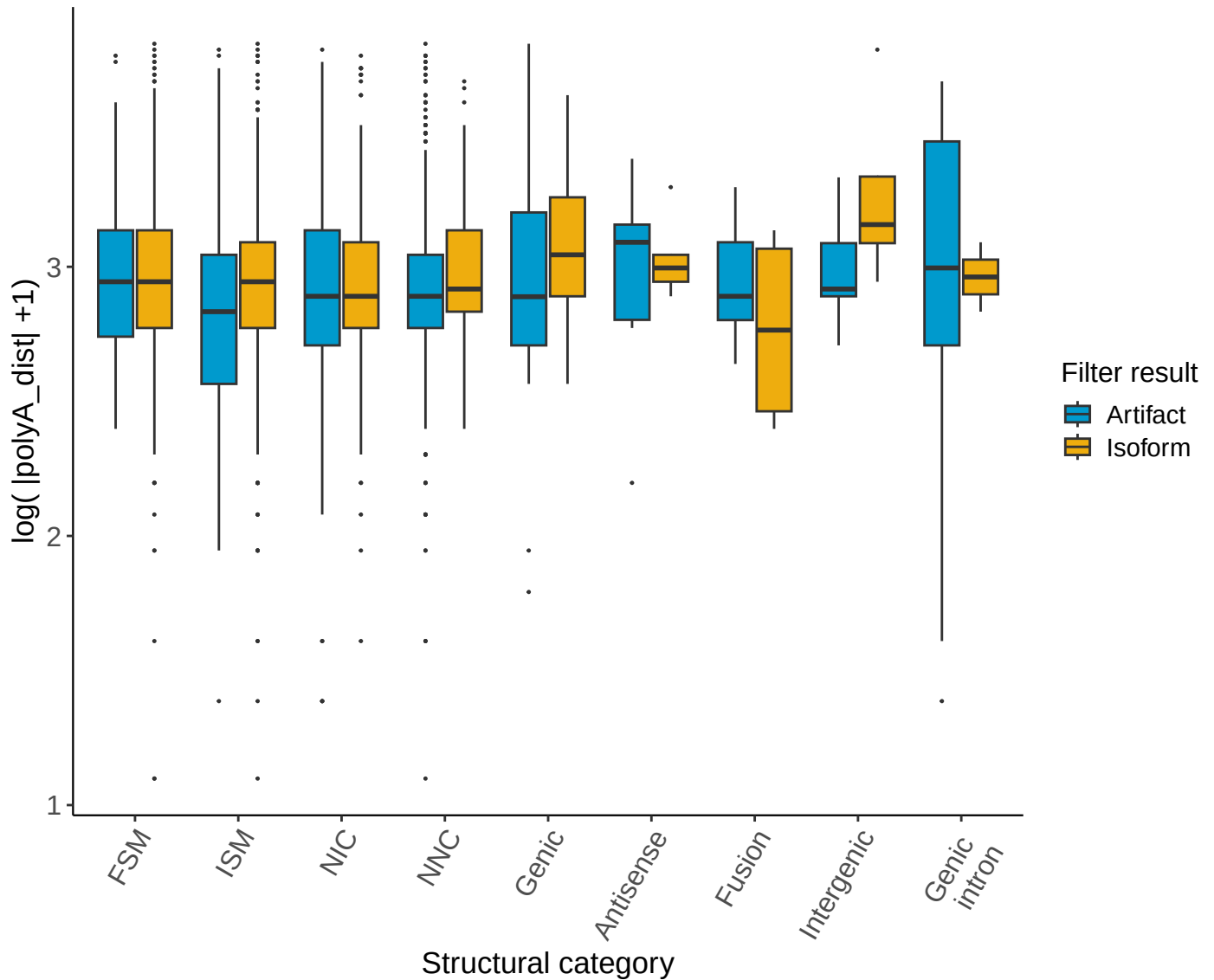
exons – ML importance: 5.31



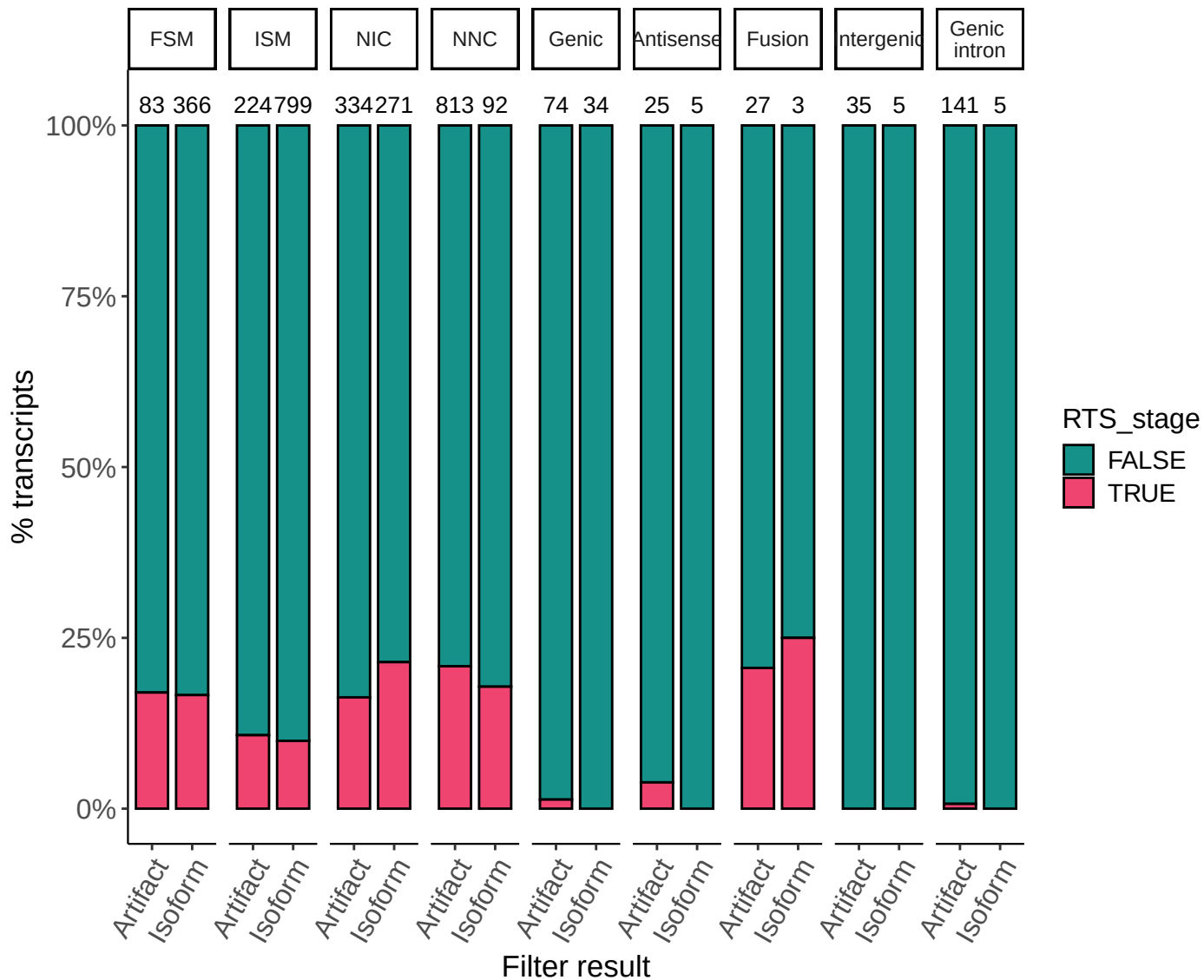
perc_A_downstream_TTS – ML importance: 4.79



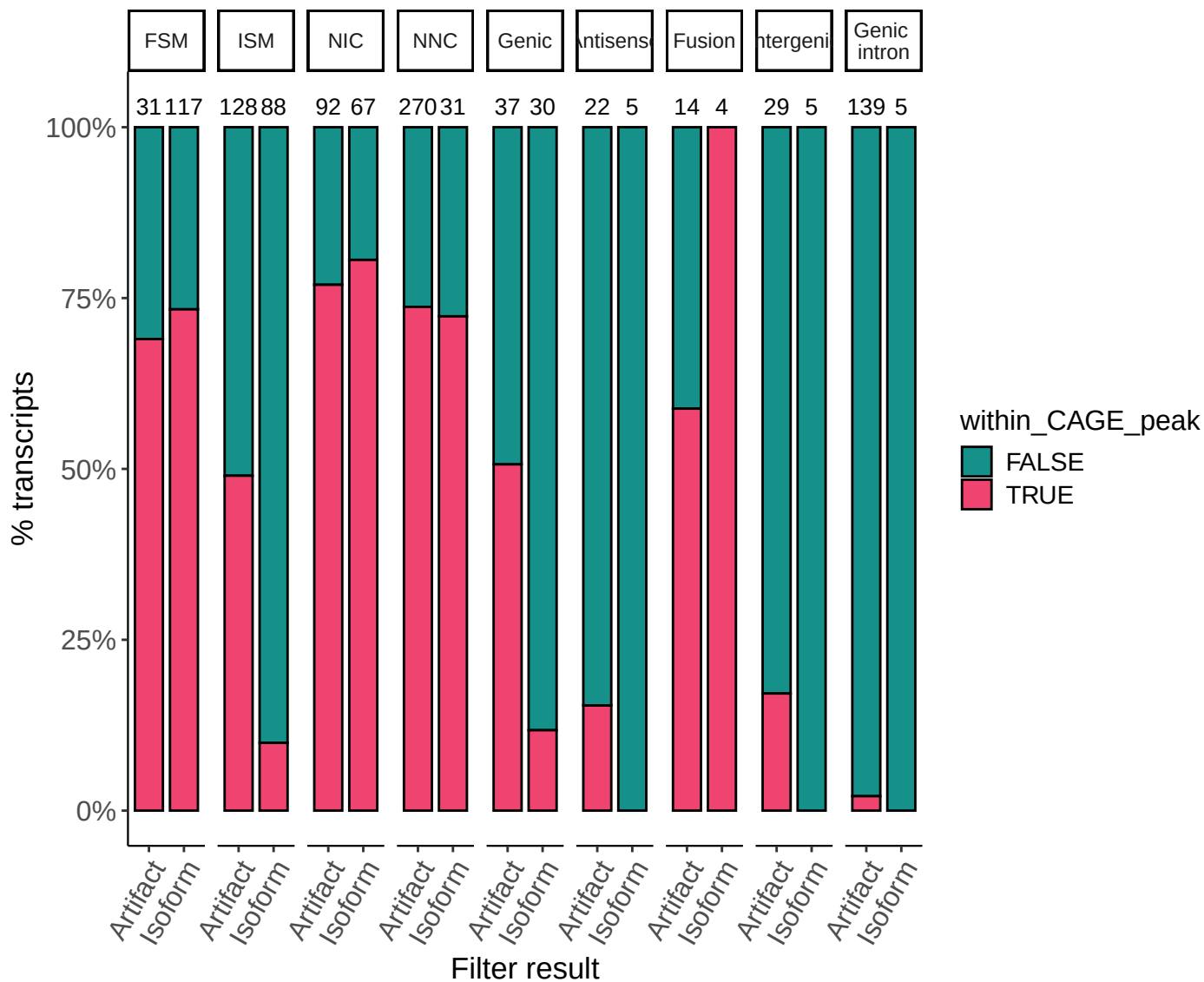
polyA_dist – ML importance: 4.72



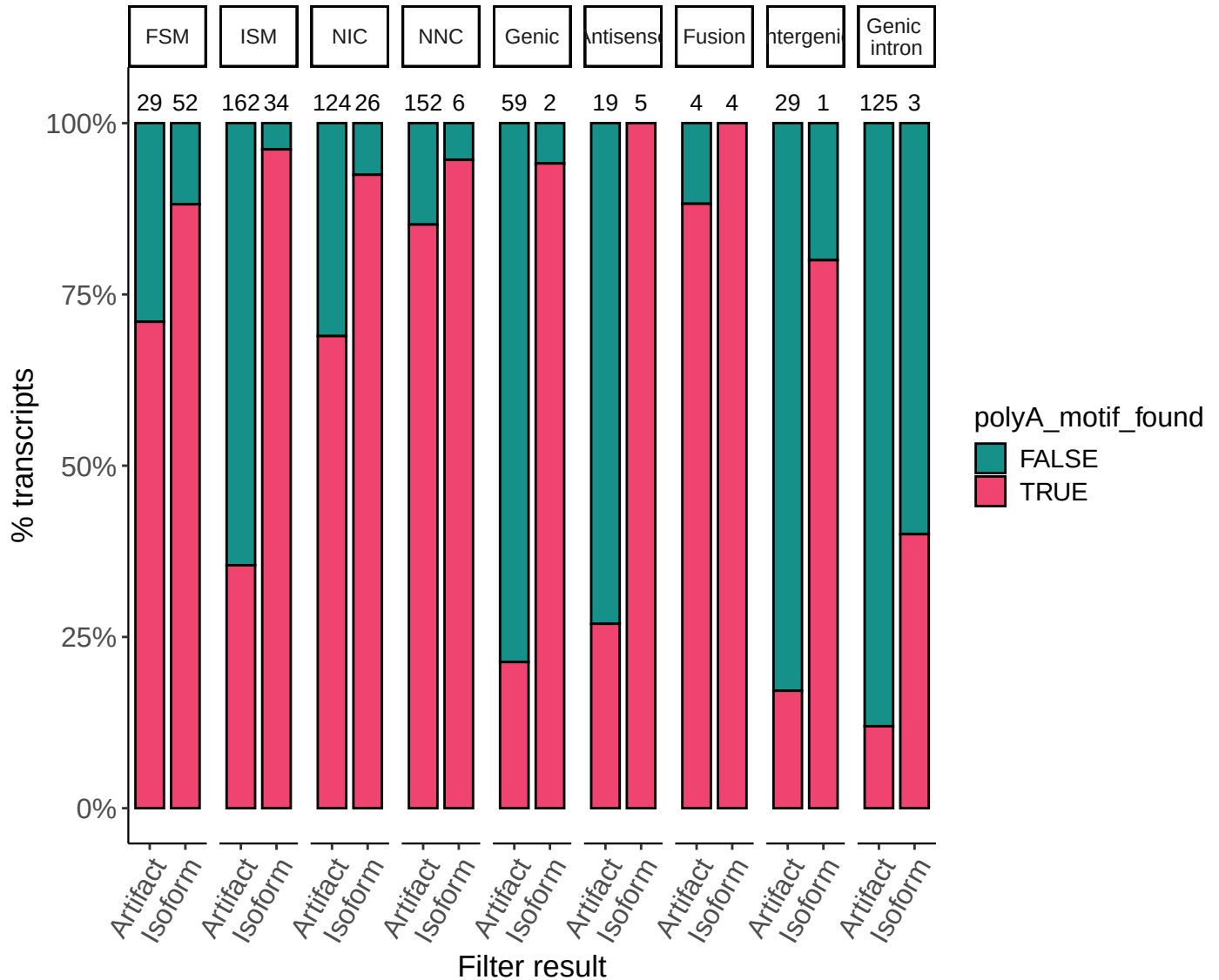
RTS_stage – ML importance: 0.93



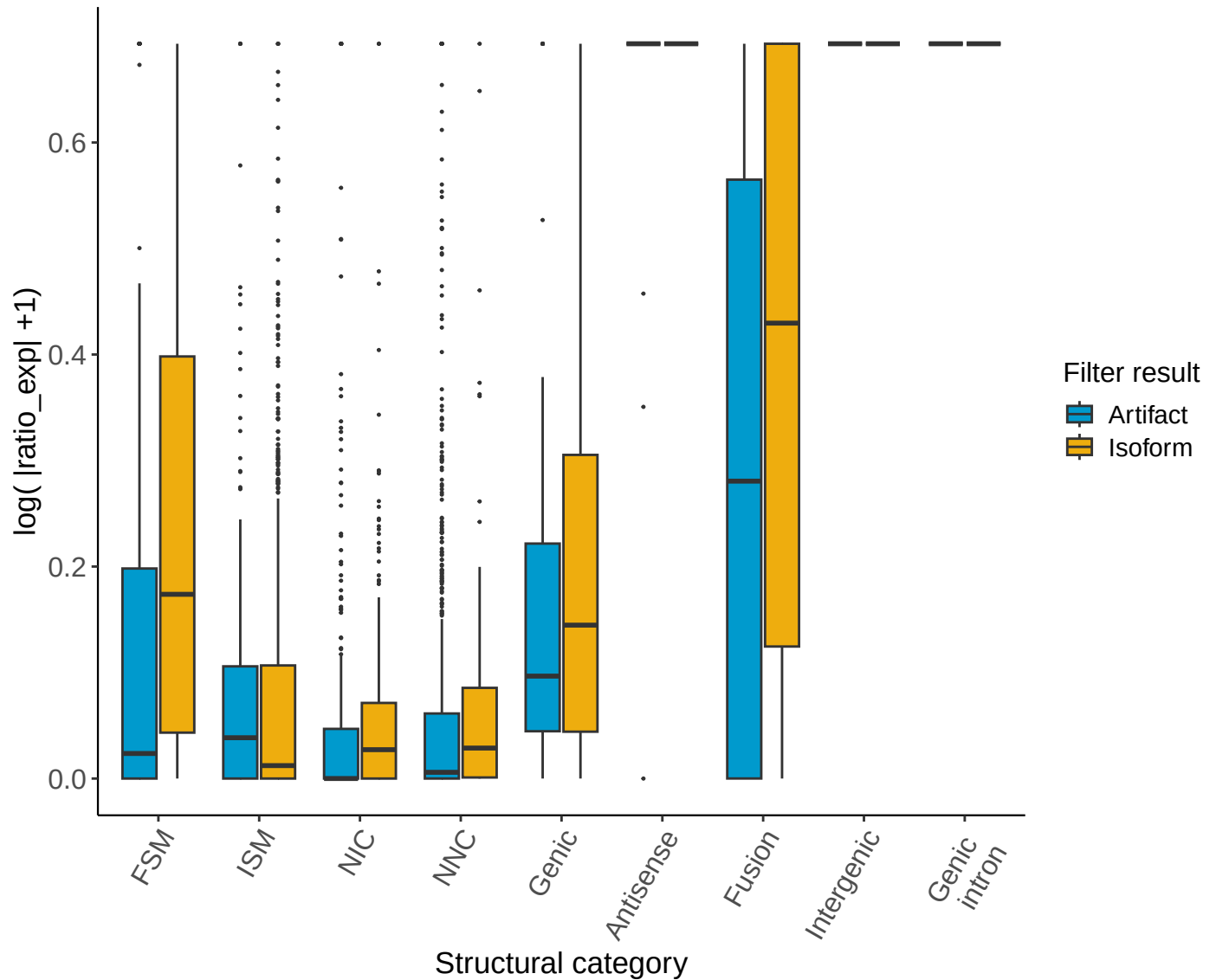
within_CAGE_peak – ML importance: 0.77



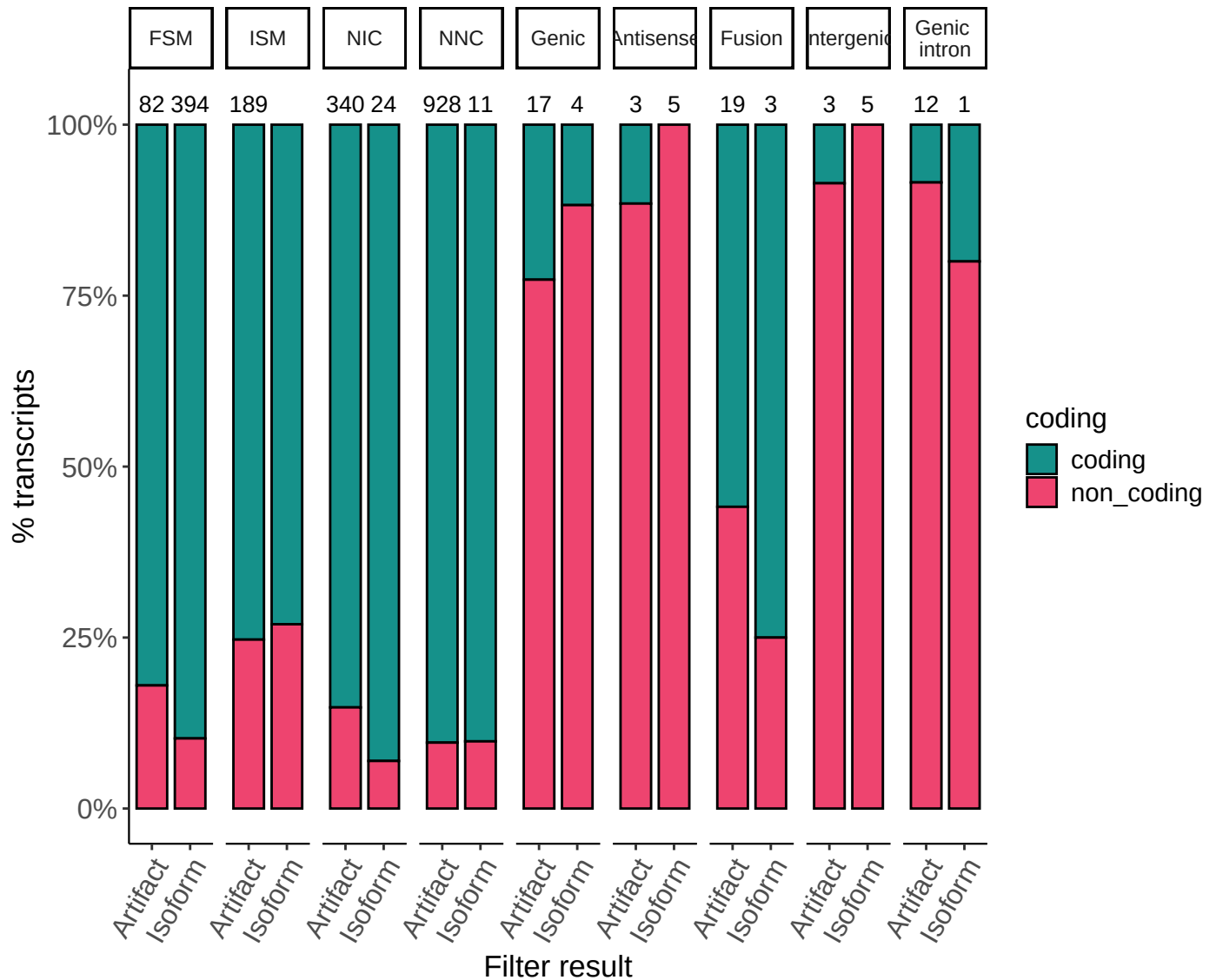
polyA_motif_found – ML importance: 0.61



ratio_exp – ML importance: 0.58



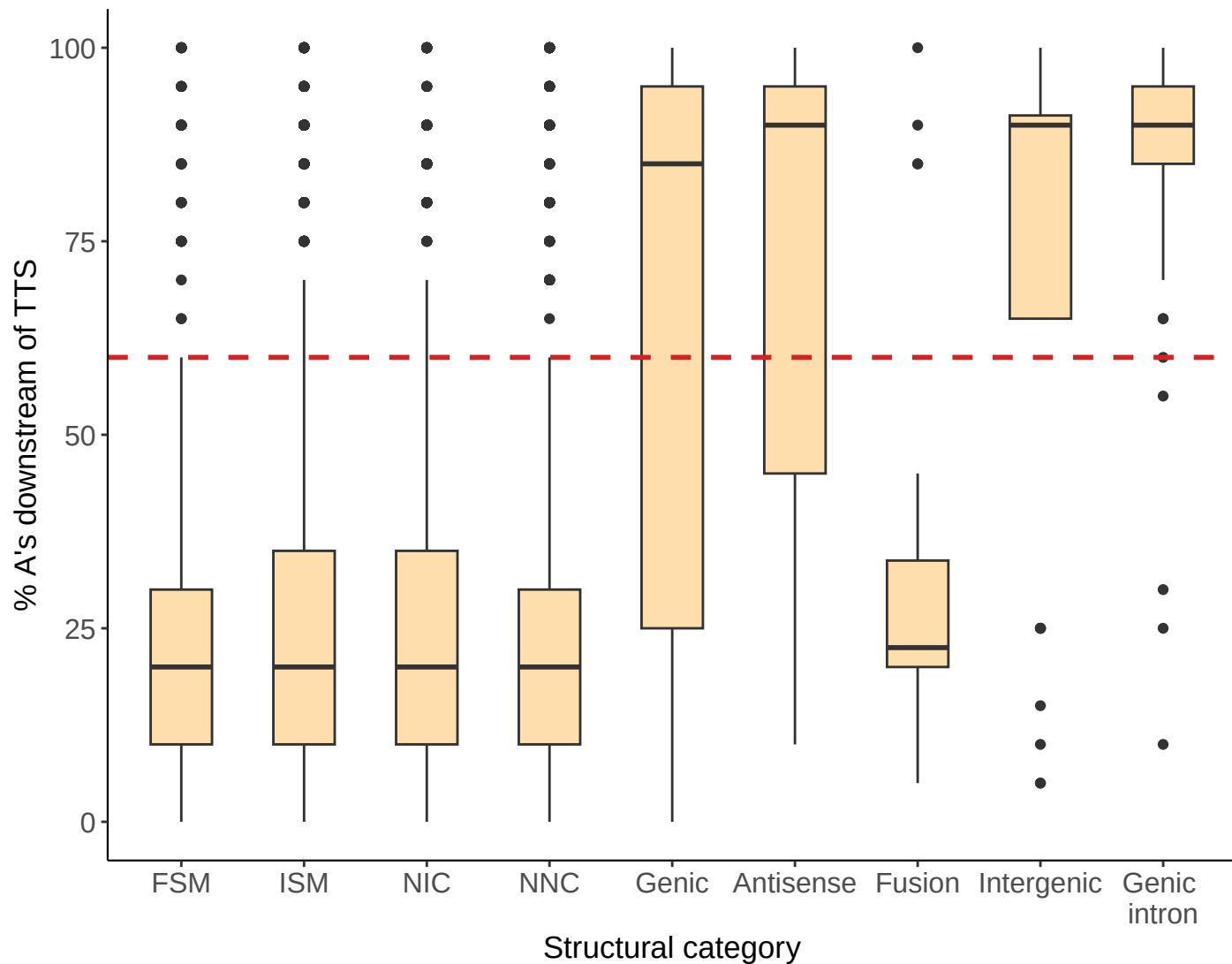
coding – ML importance: 0.24



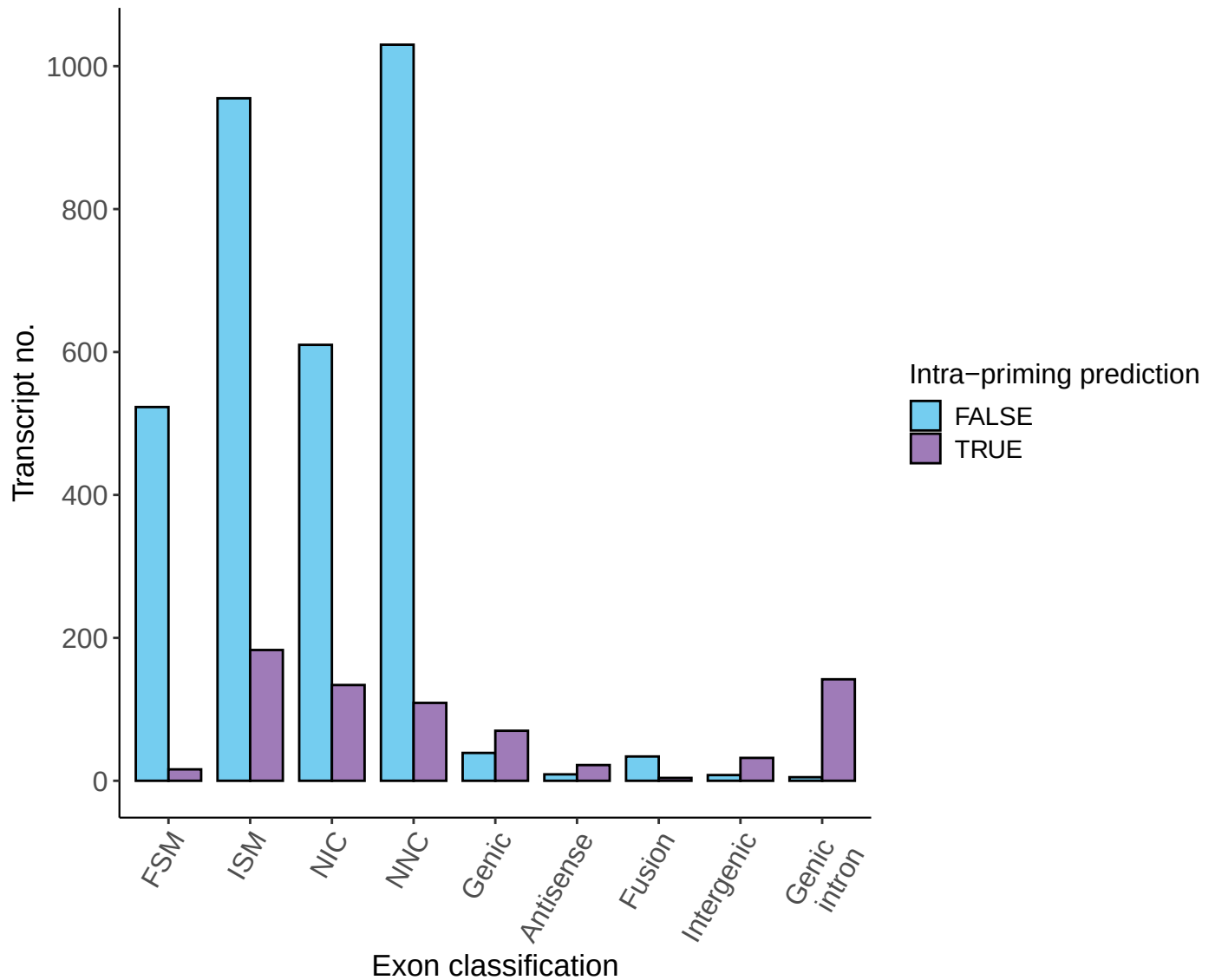
Intra-primming filter report

A % by category

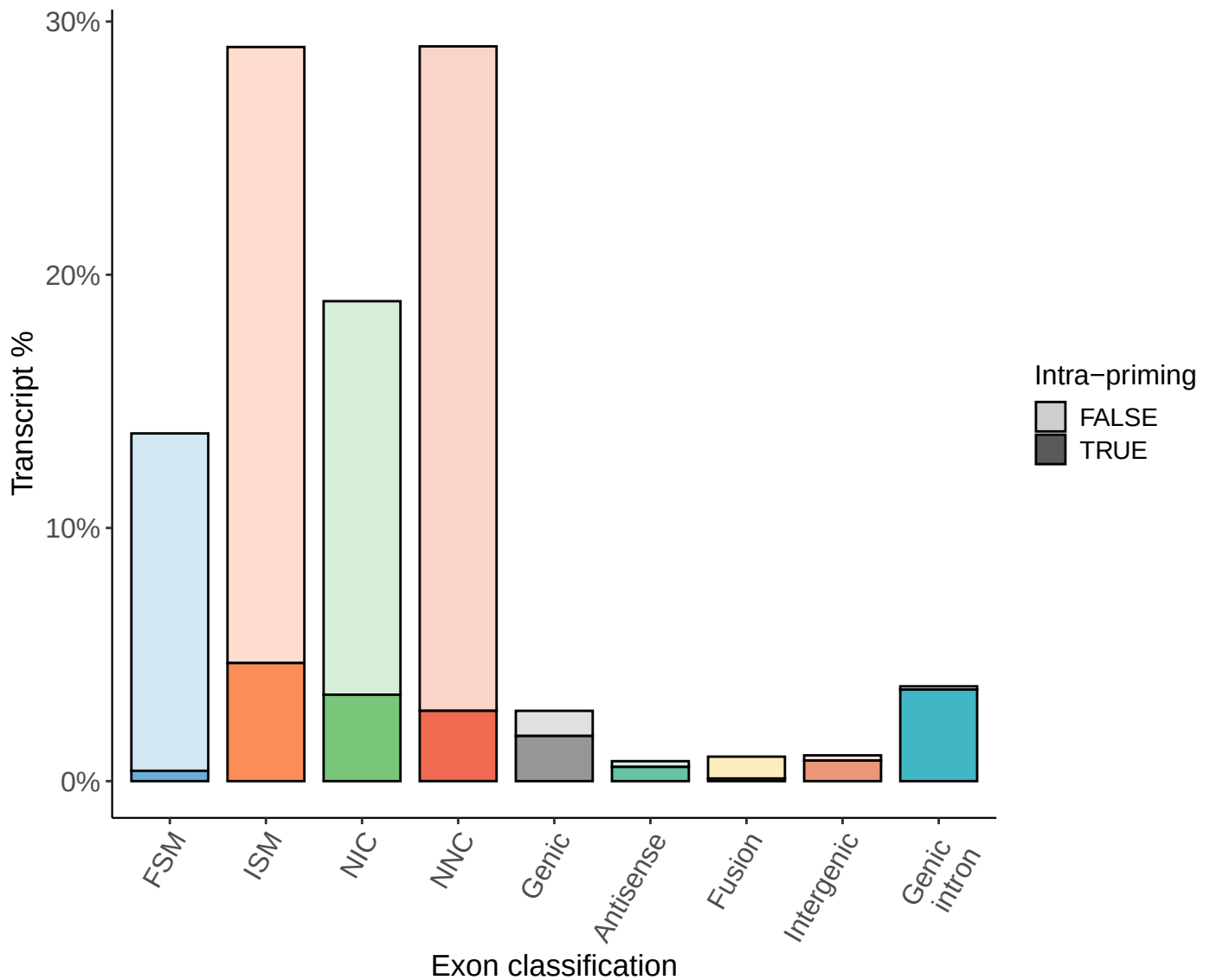
Red line indicates threshold employed in ML filter



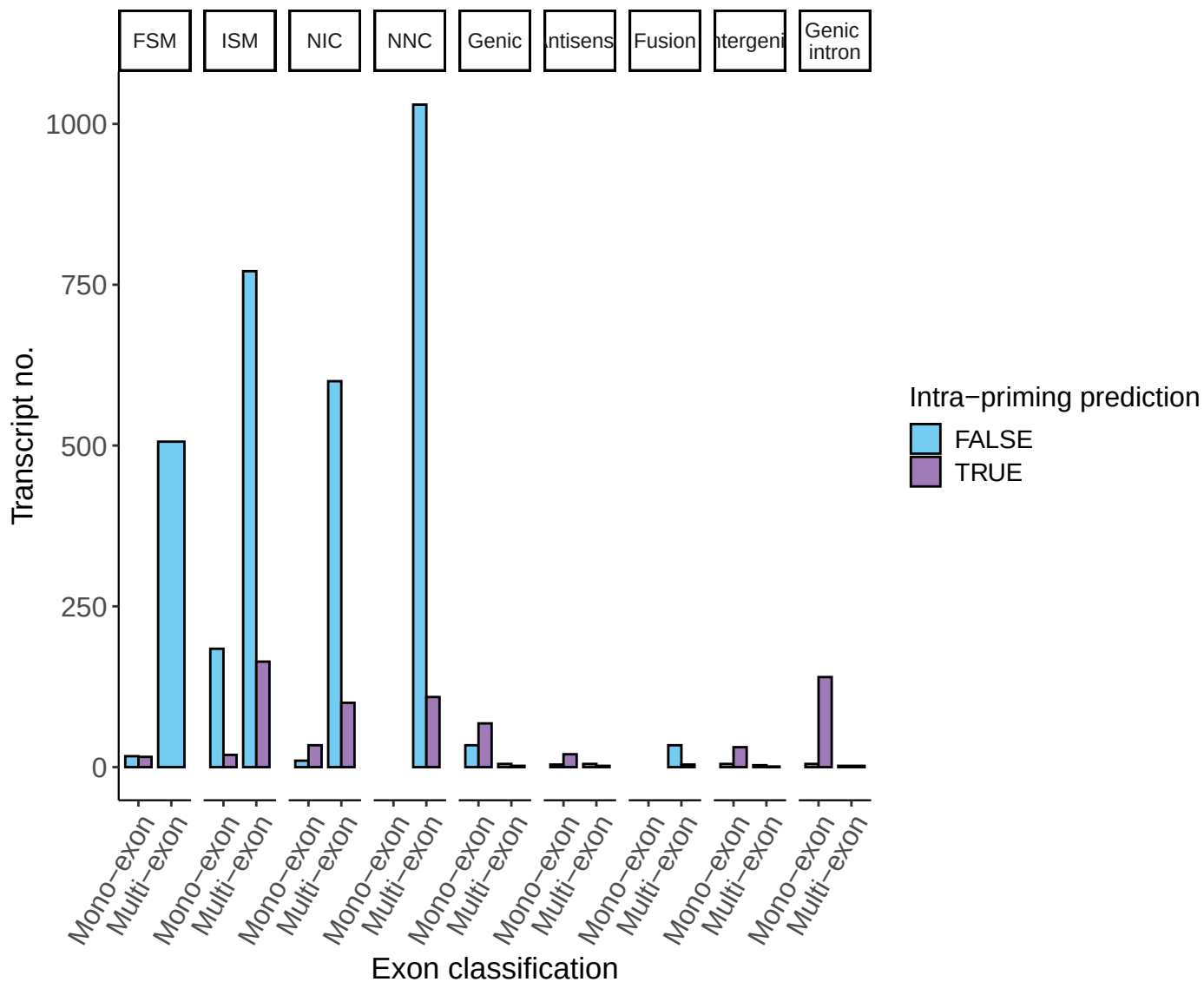
Isoforms flagged as intra-priming, by category



Isoforms flagged as intra-priming, by category (%)



Isoforms flagged as intra-priming, by exon number



Isoforms flagged as intra-priming, by exon number (%)

